CORE MATHEMATICS GRADE 10: QUADRATIC EXPRESSIONS AND EQUATIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Forming quadratic expressions from real-life and geometric contexts – Deriving quadratic identities from the concept of area (e.g., tiled maize-store floors, shamba plots) – Expanding quadratic expressions: $(a + b)^2$, $(a - b)^2$, $(a + b)(a - b)$ – Factorising quadratic expressions: common factor, grouping, trinomial method, difference of two squares – Forming quadratic equations from word problems set in Kenyan contexts – Solving quadratic equations by factorisation – Solving quadratic equations by completing the square – Solving quadratic equations using the quadratic formula – Applying quadratic equations to real-life situations (e.g., projectile motion of a thrown ball, area of a rectangular shamba, profit/loss models for a Nairobi market trader) – Interpreting solutions of quadratic equations in context
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) form quadratic expressions from real-life and geometric situations, b) derive quadratic identities from the concept of area, c) factorise quadratic expressions, d) form, solve, and apply quadratic equations in real-life situations, e) appreciate the relevance of quadratic expressions and equations in solving everyday problems.
Core Competencies	– Critical Thinking and Problem Solving: the learner forms and solves quadratic equations arising from real-life Kenyan contexts such as calculating the dimensions of a rectangular shamba in Rift Valley or maximising profit for a githeri vendor in Kisumu market. – Communication and Collaboration: the learner discusses strategies for factorising quadratic expressions and presents solutions to peers, justifying each algebraic step clearly. – Creativity and Imagination: the learner derives quadratic identities by constructing and dissecting area models of square and rectangular plots, connecting geometric reasoning to algebraic identity. – Self-Efficacy: the learner builds confidence by successfully applying multiple solution methods (factorisation, completing the square, quadratic formula) and selecting the most efficient method for a given problem. – Digital Literacy: the learner uses calculators and available digital devices to verify solutions of quadratic equations and explore

	graphical representations of quadratic functions.
Core Values	– Integrity: the learner records workings accurately and honestly checks solutions by substituting back into the original equation, modelling the precision required in engineering and financial contexts. – Responsibility: the learner takes ownership of their learning by practising multiple factorisation techniques and seeking help when stuck, rather than copying peers' work. – Respect: the learner listens to and acknowledges peers' alternative methods of solving quadratic equations during group discussions, recognising that more than one valid approach may exist. – Unity: the learner collaborates in groups to derive area-based quadratic identities, sharing materials such as graph paper cut-outs representing shamba plots, and valuing each member's contribution.
Science & Engineering Practices	– Asking Questions and Defining Problems: the learner identifies a real-life situation (e.g., a farmer in Kericho wants to fence a rectangular tea plot of known area) and translates it into a quadratic equation, articulating exactly what unknown must be found. – Using Mathematics and Computational Thinking: the learner applies algebraic manipulation — expanding, factorising, completing the square, and using the quadratic formula — as systematic computational tools for solving problems. – Developing and Using Models: the learner constructs area models on squared paper to derive the identities $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)(a + b) = a^2 - b^2$, connecting visual geometric models to symbolic algebraic expressions. – Constructing Explanations: the learner explains why a quadratic equation can have two, one, or no real solutions, linking the discriminant to the physical feasibility of solutions in context (e.g., a negative length is inadmissible for a shamba boundary). – Engaging in Argument from Evidence: the learner compares the efficiency of factorisation versus the quadratic formula for specific equations, justifying the choice of method with mathematical reasoning.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development (Financial Literacy): the learner models profit and cost scenarios for small-scale traders in Nairobi or Mombasa using quadratic equations, developing awareness of how mathematics supports sound economic decision-making. – Social Cohesion: the learner works collaboratively in mixed-ability groups to derive identities and solve problems, fostering teamwork and mutual respect across diverse learner backgrounds. – Environmental Education: the learner applies quadratic equations to land-use problems — such as determining the optimal dimensions of a rectangular shamba in the Rift Valley to maximise crop yield for a given fencing length — connecting mathematics to responsible land stewardship. – Life Skills (Decision Making): the learner interprets both roots of a quadratic equation in context, deciding which solution is physically meaningful, thereby practising evidence-based decision making.
Career Connections	– Engineering (Civil, Structural, Mechanical): quadratic equations are used to calculate beam loads, projectile trajectories, and structural dimensions — directly relevant to Kenya's ongoing infrastructure development (roads, bridges, SGR expansion). – Architecture and Construction: architects and quantity surveyors use quadratic expressions when computing areas and optimising building footprints for estates in Nairobi and other urban centres. – Agriculture and Land Management: agronomists and farm managers in regions like Nakuru and Eldoret use quadratic models to maximise yield areas given fixed fencing or irrigation resources. – Economics and Business: financial analysts and market traders use quadratic revenue and cost models to determine break-even points and profit-maximising output levels. – Physics and Data Science: physicists studying projectile motion of a thrown javelin or football, and data scientists fitting curves to datasets, rely on quadratic functions as foundational tools. – Computer Science: programmers and game developers use quadratic equations in collision detection, animation paths, and algorithm complexity analysis.

Focus for Lessons	This unit develops learners' ability to work fluently with quadratic expressions and equations — forming them from geometric and real-life Kenyan contexts, deriving identities through area models, factorising, and solving by three methods (factorisation, completing the square, and the quadratic formula). The pedagogical approach moves from concrete visual models (squared-paper shamba plots) to symbolic manipulation to real-world application, ensuring learners see quadratic algebra not as abstract symbol-pushing but as a powerful tool for answering genuine questions about land, trade, and motion in Kenya. Throughout all eight lessons the anchoring phenomenon — the puzzling parabolic path of a ball thrown by an athlete on a Nairobi school field — ties each algebraic technique back to a tangible, observable event.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems — from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho?" KICD KEY INQUIRY QUESTIONS: 1. How do we form quadratic expressions? 2. How do we factorise quadratic expressions? 3. How do we form and solve quadratic equations in real-life situations?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Parabolic Throw: During a school sports day at a secondary school in Nairobi, a student athlete throws a shot put. A classmate filming on a phone notices something striking when they replay the video in slow motion: the ball does not travel in a straight line — it curves upward, reaches a highest point, then curves back down to the ground, tracing a smooth, symmetrical arch. The class measures (using string on a printed still-frame) that the ball's height above the ground at different horizontal distances follows a curved pattern that is neither linear nor random. What is puzzling: the path looks perfectly smooth and predictable, yet the student cannot explain it with any formula learned so far. Linear equations produce straight lines; this is clearly not straight. The teacher asks: "What kind of mathematical rule could describe this curved path — and could that same rule help us figure out exactly where the ball will land before it even lands?" This puzzle — connecting a visible, real athletic event to an algebraic structure not yet encountered — anchors the entire unit and gives every technique (forming expressions, factorising, solving equations) a purpose: to decode and predict the behaviour of this curve.
Storyline Thread	Lesson 1 — Noticing the Curve / Forming Quadratic Expressions: Students watch the slow-motion throw video and trace the ball's path on printed grid paper. Teacher asks: "What kind of rule makes this shape?" Students attempt linear fits and fail. Teacher introduces the idea of an expression with an x^2 term. Learners form quadratic expressions from area contexts (rectangular shamba diagrams) and from the shot-put height data, discovering the standard form $ax^2 + bx + c$. The DQB is started: "What makes a quadratic expression different from a linear one?" Lesson 2 — Area Models and Quadratic Identities: Using squared graph paper, learners cut and rearrange pieces representing $(a + b)^2$ and $(a - b)^2$ to derive the three standard identities geometrically. They connect each identity to the shamba-enlargement supporting phenomenon. Factual and conceptual questions are added to the DQB. Lesson 3 — Expanding Quadratic Expressions: Learners practise expanding products of two linear brackets using the grid/box method and FOIL, with all examples drawn from Kenyan contexts (dimensions of a school garden in Nakuru, price-demand relationships for a Mombasa fish seller). Common errors

	are surfaced and corrected collaboratively. Lesson 4 — Factorising Quadratic Expressions (Part 1 — Common Factor, Grouping, Difference of Two Squares): Learners reverse the expansion process. The tiled courtyard and shamba supporting phenomena are revisited: students factorise expressions that arose naturally from those contexts. Emphasis on difference of two squares as a special case of the $(a+b)(a-b)$ identity. Lesson 5 — Factorising Quadratic Expressions (Part 2 — Trinomials): Learners extend factorisation to the general trinomial $ax^2 + bx + c$ using the product-and-sum method and the splitting-the-middle-term method. Real-life quadratic expressions from the mandazi profit model and the mango-drop height model are factorised, keeping the anchoring and supporting phenomena alive. Lesson 6 — Forming and Solving Quadratic Equations by Factorisation: Learners translate word problems (shamba dimensions, break-even for the mandazi seller, landing point of the shot put) into quadratic equations, set them equal to zero, factorise, and solve. They interpret both roots in context, discussing why one root may be inadmissible (e.g., negative length). DQB is updated: "Why does a quadratic equation have two solutions — and are both always meaningful?" Lesson 7 — Solving Quadratic Equations by Completing the Square and the Quadratic Formula: Learners discover completing the square by rearranging area models that cannot be factorised easily, then derive the quadratic formula from the general form. They use both methods on the shot-put parabola problem, verifying their answers with calculators. The formula is connected back to the discriminant: what does it mean for the ball's path if $b^2 - 4ac < 0$? Lesson 8 — Applications, Consolidation, and Model Revision: Learners tackle multi-step real-life problems integrating all techniques: a Rift Valley farmer optimising a fenced plot, a physics problem about a ball's flight time, and a business problem for a Nairobi market trader. Groups present solutions and revise their cumulative models on the DQB, answering the driving question with evidence from across all eight lessons. Assessment task introduced: a structured problem set and a short project connecting quadratic equations to one chosen real-life Kenyan context.
Supporting Phenomena	– The Shrinking and Growing Shamba: A farmer in Kericho has a rectangular tea plot. She increases the length by a certain number of metres and decreases the width by the same number. Students discover that the resulting area can be expressed as a quadratic expression — and that the identity $(a + b)(a - b) = a^2 - b^2$ neatly describes the relationship. Why does changing dimensions this way always produce an expression with no middle term? – The Tiled Courtyard Puzzle: A school in Kisumu has a square courtyard of side length x metres, surrounded by a uniform border of decorative tiles. The total tiled area is described by a quadratic expression. Students are shown a diagram and asked: given the total tiled area, what is the side length of the inner courtyard? Solving this requires forming and solving a quadratic equation. – The Market Profit Model: A mandazi seller at Gikomba market in Nairobi finds that her daily profit (in Ksh) depends on the number of extra mandazi she bakes beyond her base amount, described by a quadratic relationship. At what production level does she break even, and at what level is profit zero again? Students must form and solve a quadratic equation to answer a real business question. – The Falling Mango: A mango is dropped from the top of a tree on a farm in Kisii. Its height above the ground at time t seconds is modelled by a quadratic expression in t. Students observe that the mango's height decreases in an accelerating, non-linear way — and must find when it hits the ground by solving a quadratic equation, choosing the positive root as the physically

	meaningful solution.

LESSON 1 (80 minutes): Noticing the Curve — Introducing Quadratic Expressions Through the Shot Put Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon (the parabolic path of a shot put thrown at a Nairobi school sports day) and build the foundational understanding that some real-world curves cannot be described by linear equations — motivating the need for quadratic expressions.
Knowledge	Learners identify that a curved path (parabola) differs from a straight-line path; recognise that substituting values of a variable into an expression of the form $ax^2 + bx + c$ produces outputs that match a curved data pattern; and understand that a quadratic expression contains a squared variable term as its highest power.
Skills	Learners collect and tabulate height-vs-distance data from a printed still-frame image; plot points on squared paper; compare the shape of their plot to a straight-line graph; and begin to form simple quadratic expressions by expanding products of two linear brackets.
Attitudes	Learners appreciate that mathematics is a tool for explaining real, observable physical events in their environment; show curiosity and persistence when a familiar algebraic toolkit (linear equations) proves insufficient; and demonstrate respect for peers' predictions during whole-class discussion.
Key Inquiry Question	How do we form quadratic expressions?
Purpose in Storyline	This is Lesson 1 of the unit. It opens the anchoring phenomenon — the parabolic arc of a shot put thrown on a Nairobi school sports day — and establishes the central puzzle: linear equations cannot describe this curve. Every subsequent lesson technique (forming expressions, factorising, solving equations) will be motivated by the need to decode and predict this curve. This lesson plants the seed for the entire Driving Question Board and returns to the phenomenon at every phase.
Safety Notes	No physical hazards anticipated. If learners are asked to throw a small ball outdoors to replicate the phenomenon, ensure a clear open space away from windows and other learners. Supervise closely and remind learners not to throw toward people.

B. LESSON OVERVIEW
Learners are introduced to the anchoring phenomenon: a slow-motion video still of a shot put thrown at a Nairobi secondary school sports day. They first predict the shape of the ball's path using only prior knowledge, then observe tabulated height-vs-distance data and plot points on squared paper. Through guided sensemaking, they discover that the data produces a smooth curve — not a straight line — and that no linear equation $y = mx + c$ can fit it. The teacher guides learners to notice that the outputs change in a non-constant way and introduces the idea that a squared term (x^2) might be needed. Learners begin forming simple quadratic expressions by expanding pairs of linear brackets (e.g. $(x + 3)(x + 2)$) and connecting this algebraic structure to the shape of the curve. The lesson closes by opening the class Driving Question Board (DQB) with learner-generated questions, and learners draw their first rough conceptual model of the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Basic Parabola](#)	Learners are shown a printed A4 still-frame image (or projected image) taken from a phone video	Display the still-frame image on the board or projector. Say: 'Look carefully at this image. This was	Think-Pair-Share with public prediction display. This activates prior knowledge of motion and	Circulate during individual sketching. Look for: (1) learners who draw a straight diagonal line

Graph Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	of a student athlete at a Nairobi school sports day, frozen at the moment the shot put is at its highest point. The ball's starting position and a few intermediate positions are marked with dots. Learners individually sketch in their exercise books what they think the COMPLETE path of the ball looks like — from the moment it leaves the hand to the moment it hits the ground — and write one sentence explaining their reasoning. They then share predictions in pairs (Think-Pair-Share) and the class hears 4–5 different predictions read aloud.	taken at our school sports day — a student, just like you, threw a shot put. The dots show where the ball was at different moments. In your exercise book, sketch what you think the FULL path of the ball looks like from throw to landing. Then write one sentence: why do you think it follows that path?' Allow 3 minutes of silent individual work. After 3 minutes, say: 'Now share your sketch with your partner. You have 90 seconds.' After pair sharing, cold-call 4 learners (choose a mix of confident and less-confident): 'Atieno, show us your sketch and explain your thinking.' After each share, ask the class: 'Raise your hand if your sketch looks similar to Atieno's.' Record the 3 most common prediction shapes on the board labelled P1, P2, P3. Do NOT confirm or correct any prediction. Say: 'We are going to test all of these predictions today.'	straight-line graphs, surfaces misconceptions (many learners will predict a straight diagonal line or a random curve), and creates cognitive dissonance that the lesson resolves. Publicly recording predictions makes learners personally invested in finding out who was right.	— these learners are anchored to linear thinking and will need targeted questioning during Phase 3; (2) learners who draw a rough arch — these have intuitive knowledge that can be built on. Note 2–3 names from each group for strategic cold-calling later. Do not give feedback yet.
Observe Phase 📹 **VIDEO:** Basic Parabola Graph Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a data table (printed or copied from the board) showing the horizontal distance x (in metres) of the shot put from the throwing circle and the corresponding height h (in metres) above the ground, measured by a classmate using string on a printed still-frame. The table contains 7 data points: (0, 1.5), (2, 3.3), (4, 4.3), (6, 4.5), (8, 3.9), (10, 2.5), (12, 0.3). Working in groups of 3–4, learners: (a) copy the table into their exercise books, (b) plot the 7 points on a squared-paper grid (x-axis: horizontal distance 0–14 m; y-axis: height 0–6 m), (c) attempt to draw a straight line through the points using a ruler, and (d) write an observation sentence: 'A straight line (does / does not) fit all	Distribute the data table and squared paper. Say: 'This table shows real measurements a student took from the shot put video — horizontal distance along the ground and height of the ball. Your job: plot these points carefully, then try to draw a straight line through ALL of them using your ruler.' Allow 12 minutes for plotting and the straight-line attempt. Circulate — when a group correctly plots the points and sees the curve, ask: 'Can you get your ruler to pass through ALL seven points at once?' [WAIT 10 seconds]. After plotting, say: 'Write one sentence starting with: A straight line does/does not fit all the points because…' After 2 minutes of writing, say: 'Leave	Data plotting and straight-line falsification. By physically attempting to lay a ruler through all 7 points and failing, learners generate their own evidence that linearity is insufficient. The gallery walk adds a social layer — learners validate each other's findings, building collective confidence in the observation. Connecting back to the phenomenon: these are REAL measurements from the shot put arc.	Check plotted graphs during circulation: (1) Are points plotted accurately? Common error — learners reverse x and h axes. Correct immediately by asking 'Which axis shows how far the ball has travelled sideways?' (2) Is the straight-line attempt genuine? Some learners may force the ruler through only 2–3 points and claim it works — ask 'Does your line pass through the point at x = 6?' (3) Listen to sentence writing — learners who write 'does fit' need a direct probe: 'Show me which straight line passes through x=0 AND x=6 AND x=12 at the same time.'

	the points because…'. Groups then rotate and observe two other groups' graphs, adding a sticky-note comment.	your graph on your desk and walk to look at two other groups' graphs. Leave a sticky note with ONE observation about their graph.' After gallery walk (4 minutes), reconvene. Cold-call 3 groups: 'Kamau's group — what did you write in your sentence?' After 3 responses, say: 'Nearly every group found the same thing. What does that tell us about this path?' [WAIT 15 seconds]. Accept 2–3 responses. Write on the board: 'The path is NOT a straight line → a linear equation $y = mx + c$ CANNOT describe it.'		
Explain Phase 📹 **VIDEO:** Basic Parabola Graph Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class discussion connecting the curved shape to algebra. Learners first compare the differences in the height values (first differences) and then the differences of those differences (second differences), discovering that second differences are approximately constant — a hallmark of quadratic relationships. The teacher then introduces the general quadratic expression $ax^2 + bx + c$ and demonstrates, with learner participation, that substituting $x = 0, 2, 4, 6$ into $h = -0.1x^2 + 1.2x + 1.5$ produces values close to the observed data. Learners then practise forming quadratic expressions by expanding two linear brackets: $(x + 3)(x + 2)$, $(x + 5)(x - 1)$, and $(2x + 1)(x + 4)$, recording each step in their exercise books. They connect this to the phenomenon: 'The expression describing the shot put's path could be written as a product of two simpler expressions — which we will learn to find later in this unit.'	Write the height data column on the board: 1.5, 3.3, 4.3, 4.5, 3.9, 2.5, 0.3. Say: 'Let us look at how the height changes each time x increases by 2 metres.' Calculate first differences with the class by cold-calling: 'Wanjiku, what is 3.3 minus 1.5?' [WAIT 10 seconds]. Record: +1.8, +1.0, +0.2, −0.6, −1.4, −2.2. Say: 'Are these first differences constant?' [WAIT 10 seconds]. Cold-call: 'Ochieng, what do you notice?' Then say: 'Now let us find the differences OF those differences.' Calculate second differences: −0.8, −0.8, −0.8, −0.8, −0.8. Ask: 'What do you notice NOW?' [WAIT 15 seconds]. Cold-call 3 learners. Write on board: 'CONSTANT SECOND DIFFERENCES → QUADRATIC RELATIONSHIP.' Say: 'This is the signature of a quadratic expression. It contains an x^2 term.' Write: $h = ax^2 + bx + c$. Say: 'Someone has already worked out that $a = -0.1$, $b = 1.2$, $c = 1.5$ fits our data. Let us check together.' Substitute $x = 0$: $h = 0 + 0 + 1.5 = 1.5$ ✓. Ask a learner to substitute $x = 6$ at the board. Then	Second-difference analysis followed by bracket expansion. The second-difference pattern is a powerful, accessible sensemaking anchor — it requires only subtraction yet reveals the deep algebraic structure. Connecting the expansion of brackets to the curved data ties the abstract skill to the concrete phenomenon, preventing procedural learning in isolation. This is the NGSS Science and Engineering Practice of 'Analysing and Interpreting Data' applied in a mathematics context.	During the second-difference activity, listen for learners who compute differences incorrectly (negative numbers cause errors — watch for sign mistakes). During bracket expansion, circulate and check: (1) Does the learner correctly apply the distributive property to all four terms? (2) Does the learner correctly identify x^2 as the highest power? Use mini-whiteboards or rough work on desks — ask learners to hold up their expanded form of $(x + 5)(x - 1)$ before moving on. If more than 30% have errors, pause and reteach the distributive step using the area model (draw a rectangle divided into 4 parts).

		say: 'Now — where does this expression come from? One way: it can be formed by multiplying two linear brackets.' Write $(x + 3)(x + 2)$ on the board. Say: 'Expand this — multiply each term in the first bracket by each term in the second. Work individually for 2 minutes.' After 2 minutes, cold-call one learner to write the answer on the board. Repeat for $(x + 5)(x - 1)$ and $(2x + 1)(x + 4)$. After each, ask: 'What is the highest power of x in your answer?' Establish that it is always x^2. Close by saying: 'Every quadratic expression has x^2 as its highest power. That squared term is what creates the CURVE we saw in the shot put path.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Basic Parabola Graph] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the FIRST sticky note (yellow), they write one question they still have about the shot put curve or about quadratic expressions — something they do not yet understand. On the SECOND sticky note (blue), they write one connection they have already made — something they now understand that they did not at the start of the lesson. Learners come to the front and stick their notes on the class DQB (a large sheet of chart paper divided into two columns: 'Questions we still have' and 'Things we now understand'). The teacher reads aloud 5–6 questions, clusters similar ones, and labels 3 big question clusters that will guide the rest of the unit.	Distribute two sticky notes to each learner. Say: 'On your YELLOW note, write one question you still have — something about this curve or about quadratic expressions that you do not yet understand. Be specific. Not just 'I don't understand' — write WHAT exactly puzzles you.' Give 2 minutes of silent writing. Then: 'On your BLUE note, write one thing you NOW understand that you did NOT understand at the start of today's lesson.' Give 2 minutes. Say: 'Come up and place your yellow note in the QUESTIONS column and your blue note in the THINGS WE UNDERSTAND column.' Once all notes are placed, read aloud 5–6 yellow notes and cluster visually: group questions about 'what does the x^2 term do to the shape' together, 'how do we find a, b, c for the shot put' together, and 'how do we solve for x when h = 0 (where does the ball land?)' together. Circle the cluster about landing point and	Public knowledge-state mapping via the DQB. By externalising both questions and current understanding, the DQB creates a living record of the class's intellectual journey. Clustering learner questions around the phenomenon's key unknowns (landing point, shape of curve, how to find a/b/c) makes the unit roadmap feel learner-generated rather than teacher-imposed. This mirrors the NGSS practice of 'Asking Questions and Defining Problems.'	Read all yellow sticky notes before the next lesson to diagnose: (1) Are learners asking procedural questions ('how do I expand brackets?') or conceptual ones ('why does x^2 cause a curve?')? A class dominated by procedural questions signals that the phenomenon link is not yet made — revisit the second-difference pattern at the start of Lesson 2. (2) Are there misconceptions visible ('the ball follows a circle, not a parabola')? Address these explicitly at the start of Lesson 2. (3) Do blue notes show genuine conceptual gains or surface-level restatements? Note 3–4 blue notes that show deep understanding to celebrate publicly next lesson.

		say: 'This question — WHERE exactly will the ball land? — is one we will be able to answer by the end of this unit. It will require everything we learn: forming expressions, factorising, and solving quadratic equations.' Write the driving question at the top of the DQB: 'How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems — from the arc of a thrown ball to the dimensions of a farmer's shamba in Kericho?'		
Model Building Phase 📹 **VIDEO:** Basic Parabola Graph Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners individually draw their FIRST MODEL in their exercise books. The model must include: (a) a sketch of the shot put's parabolic path with the 7 data points marked, (b) a label showing the quadratic expression $h = -0.1x^2 + 1.2x + 1.5$ next to the curve, (c) an arrow pointing to the x^2 term with the annotation 'THIS causes the curve', (d) a question mark at the point where the ball hits the ground ($x \approx 12$ m) with the annotation 'We need to SOLVE a quadratic equation to find the exact landing point.' Learners write a 2-sentence model statement: 'My model shows… My model does NOT yet explain…'	Say: 'For the last 8 minutes, you are going to build your first model of this phenomenon. A model in mathematics is a diagram plus algebra that explains what we observe. Open a fresh page and title it: MY SHOT PUT MODEL — LESSON 1.' Write the four required model components on the board and point to each: '(a) sketch the path with the 7 data points, (b) label the expression $h = -0.1x^2 + 1.2x + 1.5$, (c) annotate the x^2 term, (d) mark the landing point with a question mark.' Give 6 minutes of individual work. Then say: 'Write two sentences at the bottom: My model shows… My model does NOT yet explain…' Give 2 minutes. Collect or photograph 4–5 models (with permission) to use as anchor examples at the start of Lesson 2. Close by saying: 'Every lesson in this unit, you will come back to this model and add to it. By the end of the unit, your model will be complete enough to predict EXACTLY where any thrown object will land — before it lands.'	Cumulative annotated model building. Requiring learners to explicitly label what the x^2 term DOES (causes the curve) and to mark the landing point as an unsolved question creates a visual and linguistic anchor for the entire unit. The 'My model does NOT yet explain…' sentence stem is a metacognitive tool that keeps learners aware of the boundary between current and future knowledge — preventing premature closure. This model will be revised at the end of each subsequent lesson.	Review the 4–5 collected models before Lesson 2. Check: (1) Is the path sketched as a smooth arch (not a triangle or zigzag)? (2) Is the expression correctly transcribed? (3) Does the annotation of x^2 show any understanding of WHY it causes a curve, or is it just copied? (4) Is the landing-point question mark placed at approximately $x = 12$ m? Models that lack the question mark suggest learners do not yet see solving quadratic equations as a necessary next step — prioritise this connection in Lesson 2's opening.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions: (1) Did the prediction phase surface the expected misconception that the shot put path is a straight line? If most learners predicted a curve, consider whether a more dramatic contrast (e.g. showing a linear distance-time graph side by side) is needed in future to make the non-linearity more striking. (2) During the second-difference activity, were learners able to compute differences of negative numbers confidently? If sign errors were widespread, plan a 5-minute warm-up on directed number arithmetic for Lesson 2. (3) How rich were the DQB questions? If most questions were procedural (how do I expand brackets?) rather than phenomenon-driven (how do I find where the ball lands?), revisit the phenomenon connection more explicitly and consider showing a short video of the shot put arc before Lesson 2. (4) Did the 80-minute timing work? The observe and explain phases are the most time-sensitive — if the class ran short, the model-building phase can be completed as a homework task. (5) Which learners were most engaged by the Kenyan sports context? Use their engagement to peer-tutor those who disengaged. Note the names of learners who appeared lost during bracket expansion — these learners may need a brief prior-knowledge intervention on the distributive property before Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that when the height and horizontal distance measurements from the shot put video are plotted on squared paper, the 7 data points form a smooth arch that cannot be fitted by a straight line — falsifying the linear model $y = mx + c$.
What did I learn?	Learners learned that a quadratic expression (containing an x^2 term as its highest power) is needed to describe a curved relationship; that constant second differences in a data table signal a quadratic pattern; and that quadratic expressions can be formed by expanding products of two linear brackets.
How does this explain the phenomenon?	Learners explained (and began to model) why the shot put's path curves — the x^2 term in the expression $h = -0.1x^2 + 1.2x + 1.5$ produces outputs that increase then decrease, matching the observed rise-and-fall of the ball, unlike the constantly-changing outputs of a linear equation.

LESSON 2 (80 minutes): Forming Quadratic Expressions: From the Shot Put Arc to the Shamba	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to form quadratic expressions from real-life contexts and identify the standard form $ax^2 + bx + c$, connecting the algebra to the curved path of a thrown shot put and the area of a rectangular shamba.
Knowledge	Learners will know: (a) the meaning and structure of a quadratic expression in the form $ax^2 + bx + c$; (b) how quadratic expressions differ from linear expressions; (c) how to form a quadratic expression from an area context and from a height-distance data pattern.
Skills	Learners will be able to: (a) correctly form quadratic expressions from area and motion contexts; (b) identify the coefficients a, b, and c in a quadratic expression; (c) distinguish between linear and quadratic expressions by analysing degree and shape.
Attitudes	Learners will appreciate that algebraic expressions are tools for modelling real physical events — such as the trajectory of a shot put at a Nairobi school sports day — and that quadratic expressions are particularly powerful for describing curved, non-linear phenomena in their environment.
Key Inquiry Question	How do we form quadratic expressions?
Purpose in Storyline	In Lesson 1, learners encountered the anchoring phenomenon — the parabolic arc of a shot put — and confirmed that linear equations cannot model the curved path. In this lesson, learners take the next step: they form quadratic expressions from two concrete Kenyan contexts (a rectangular shamba in Kericho and the shot put height data), discover the structure $ax^2 + bx + c$, and begin to articulate what makes a quadratic expression fundamentally different from a linear one. This lesson is the algebraic foundation upon which factorisation and equation-solving will be built in subsequent lessons.
Safety Notes	No physical hazards anticipated. If learners use rulers or compasses for the shamba diagram activity, remind them to handle pointed instruments carefully. Ensure printed grid sheets are distributed before the lesson to avoid crowding at the teacher's desk.

B. LESSON OVERVIEW
Learners begin by re-examining the slow-motion shot put video still-frame and their traced curves from Lesson 1. The teacher facilitates a brief Predict phase in which learners attempt to write a linear rule for the ball's height — and confront its failure. In the Observe phase, learners collect evidence from two sources: (1) a printed height-versus-horizontal-distance data table from the shot put, and (2) a rectangular shamba diagram showing a Kericho tea farmer who wants to maximise area. In the Explain phase, learners form expressions from the shamba area context, expand them, and recognise the emerging x^2 term — arriving at the standard form $ax^2 + bx + c$. They then apply the same structure to the shot put height data. The DQB is updated with the class's first formal answer to "What makes a quadratic expression different from a linear one?" and learners revise their initial algebraic models accordingly.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Video: Area]	Learners receive a printed data table showing the shot put ball's approximate height (in metres)	Distribute the data table and grid paper. Say: 'In Lesson 1 we traced the ball's path and agreed it is not	Productive Failure — learners deliberately attempt and fail to fit a linear model, making the cognitive	Teacher scans pair-written predictions on the grid paper. Observable indicator: learner can

Model Introduction Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	above the ground at horizontal distances of 0, 1, 2, 3, 4, and 5 metres (e.g., 1.8, 3.2, 4.0, 4.2, 3.8, 2.8, 1.2 m). Working in pairs, they attempt to find a single linear rule $h = mx + c$ that fits all the data points by plotting two points and drawing a straight line on their grid paper. They then check whether the line passes through the remaining points and write a sentence explaining what they observe. Learners also write a prediction: 'I think the rule for this curve involves the term _______ because _______.'	a straight line. Today I want you to try — really try — to fit a straight line to this data. Work in pairs. You have 5 minutes.' [Allow 5 minutes of independent pair work.] Circulate and listen. Do NOT correct errors — let the failure of the linear model speak for itself. After 5 minutes, cold-call 3 pairs: 'Achieng and Mwangi — what line did you try, and did it work?' [WAIT 12 seconds after each question before nominating.] Record the three attempted linear equations on the board. Ask the whole class: 'Hands up — did ANY pair find a straight line that fits ALL six points?' [Expected: no hands.] Say: 'This failure is important. It is our evidence that we need a new kind of expression. Hold that thought.'	need for a quadratic expression felt rather than told. This strategy ensures that when the x^2 term is introduced, learners experience it as a solution to a real problem they have already struggled with, deepening retention.	articulate in writing that no straight line fits all six data points, and at least half of predictions mention 'x squared' or 'a curve' or 'a new term'. Teacher notes any pair that claims a linear fit works — these pairs receive targeted follow-up questioning during Phase 2.
Observe Phase 📹 **VIDEO:** Video: Area Model Introduction Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners engage with two evidence sources simultaneously in a jigsaw structure. Group A (half the class) works with the Shamba Diagram: a printed rectangular shamba in Kericho where one side is labelled x metres and the other side is $(x + 4)$ metres. Learners write expressions for the perimeter and the area, expand the area expression, and record what they notice about the highest power of x. Group B (half the class) works with the Shot Put Height Table: using the six data points, learners calculate the first differences (Δh per metre) and second differences ($\Delta^2 h$), recording results in a prepared two-column extension table. They notice that first differences are not constant (confirming non-linearity) but second differences are approximately constant (hinting at a quadratic). After 10 minutes,	Before distributing materials, say: 'We are going to look at two pieces of evidence — one from a shamba in Kericho, one from our shot put. Both are going to tell us the same algebraic secret. Your job is to find it.' Distribute shamba diagrams and extension data tables. After groups settle, circulate to Group B and prompt: 'Calculate h at x=1 minus h at x=0. Write that down. Now x=2 minus x=1. What do you notice about those differences?' [WAIT 10 seconds.] Cold-call 2 learners from Group B: 'Otieno — are your first differences constant or changing?' For Group A, prompt: 'Expand the brackets in your area expression. What is the highest power of x you see?' [WAIT 12 seconds.] Cold-call 2 learners from Group A: 'Wanjiku — what term appeared that was not in our linear expressions last lesson?' At the halfway mark (10	Jigsaw Evidence Gathering with Structured Observation — two independent evidence streams (area context and numerical difference analysis) converge on the same algebraic feature (the x^2 term). By experiencing both streams, learners build convergent evidence rather than accepting the teacher's word. The Evidence Capture Sheet externalises thinking and makes reasoning visible for formative assessment.	Teacher collects Evidence Capture Sheets at the end of Phase 2 (or photographs them). Observable indicators: (1) Group A learners correctly expand $(x)(x+4)$ to $x^2 + 4x$; (2) Group B learners correctly compute second differences and identify them as approximately constant; (3) Both groups use the word 'squared' or write 'x^2' in the 'What it might mean' column. Learners who write only 'it is bigger' in the meaning column are flagged for additional scaffolding in Phase 3.

	Group A and Group B swap tasks. All learners complete both evidence sources. Learners record their observations on a structured 'Evidence Capture Sheet' with three columns: What I did, What I noticed, What it might mean.	min), call a 30-second pause: 'Freeze. Everyone look at what the other group has been doing. In 10 seconds, swap your materials and continue.'		
Explain Phase 📹 **VIDEO:** [Video: Area Model Introduction] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher formally introduces the standard form of a quadratic expression: $ax^2 + bx + c$. Learners participate in a guided 'expression-building' activity using three real Kenyan contexts printed on cards: (1) A maize farm near Nakuru where the length is $(2x + 3)$ m and the width is $(x + 5)$ m — find the area expression; (2) A mandazi seller in Kisumu charges x shillings per mandazi; total revenue when selling $(x + 10)$ mandazis expressed as an expression; (3) A matatu travelling from Nairobi to Thika where distance is modelled as a function of time — given a simplified expression $3t^2 - 2t + 1$, identify a, b, and c. Learners work individually first (5 min), then compare with a partner (3 min), then share whole-class (5 min). The teacher records all three expressions on the board and annotates each term, labelling a (coefficient of x^2), b (coefficient of x), and c (constant). Learners update their Evidence Capture Sheet with a definition of a quadratic expression in their own words.	Open Phase 3 by saying: 'Every piece of evidence you collected points to one algebraic structure. Let me show you its name and shape.' Write $ax^2 + bx + c$ on the board in large, coloured chalk — use three colours for a, b, c respectively. Say: 'This is a quadratic expression. The word quadratic comes from the Latin quadratus — meaning square. The x^2 term is the reason the shot put curves.' Distribute the three context cards. Say: 'Work alone for 5 minutes. Expand and write each expression in the form $ax^2 + bx + c$. Identify a, b, and c.' [Set a visible 5-minute timer.] After 5 minutes: 'Now compare with your partner. Do your a, b, c values match? If not, find the disagreement — do not just copy.' [Allow 3 minutes.] Cold-call 3 learners (one per context card): 'Kamau — for the maize farm, what is your expression and what is a?' [WAIT 12 seconds each.] Annotate the board responses. Ask the whole class: 'What is the MOST IMPORTANT difference between this expression and a linear one like $3x + 5$?' [WAIT 15 seconds.] Accept 3 answers before confirming: 'The degree — the highest power of x. In a linear expression it is 1. In a quadratic it is 2.'	Annotated Example Analysis with Peer Comparison — learners form expressions independently, compare with a peer to surface disagreements, then validate against whole-class discussion. Colour-coded annotation of a, b, c on the board provides a persistent visual anchor. Connecting the Latin root 'quadratus' to the x^2 term builds vocabulary retention and links the algebra to the geometric idea of squaring.	Teacher checks individual written expressions on context cards before peer comparison begins (circulating, scanning for correct expansion). Observable indicators: (1) Learner correctly expands two binomials to form $x^2 + bx + c$; (2) Learner correctly identifies a, b, c for all three cards; (3) Learner's own-words definition includes 'x squared' or 'degree 2' or 'highest power is 2'. Learners who cannot expand the brackets are noted for a targeted worked example at the start of Lesson 3.
Driving Question	Each learner receives two sticky notes. On the first (green), they	Say: 'It is time to update our Driving Question Board. In Lesson	Driving Question Board — Running Record — the DQB acts	Teacher reads all green notes after the lesson. Observable

Board (DQB) Creation 📹 **VIDEO:** Video: Area Model Introduction Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	write one thing they can now answer from the DQB: 'What makes a quadratic expression different from a linear one?' — and they write their answer. On the second (orange), they write a new question that this lesson has raised for them. Learners post both notes on the class DQB. The teacher reads out 4–5 green notes and 4–5 orange notes aloud to the class. The class votes (thumbs up/sideways/down) on whether each green-note answer is correct, complete, or needs revision. The teacher annotates the DQB with a large arrow connecting today's evidence (shamba area, shot put data) to the question 'What kind of rule makes this shape?' — written in Lesson 1 — and writes the class's agreed answer beneath it: 'A quadratic expression — $ax^2 + bx + c$ — because it contains an x^2 term, giving it a curved (parabolic) shape.'	1 we asked: what kind of mathematical rule could describe this curved path? Today you have found evidence to answer part of that question.' Distribute sticky notes. Say: 'Green note — write your answer to the question on the board. Orange note — write the NEW question this lesson has raised in your mind. You have 3 minutes.' [WAIT 3 minutes.] Collect and post notes quickly. Read 4 green notes aloud, pausing after each: 'Class — thumbs up if this answer is correct and complete. Thumbs sideways if it is partly right. Thumbs down if something is missing.' [WAIT 8 seconds for thumbs after each.] After reading orange notes, say: 'These new questions — about factorising, about solving, about the shape of the graph — are exactly where we are going in the next six lessons. Pin them here. We will answer them one by one.'	as a cumulative public record of the class's growing understanding. By physically connecting today's evidence arrows to last lesson's questions, learners see their own intellectual progress. Peer voting on green-note answers builds metacognitive awareness and collective accountability for accuracy.	indicators: (1) At least 80% of green notes include 'x^2' or 'degree 2' or 'squared term' as the distinguishing feature; (2) Orange notes reveal productive next questions (e.g., 'How do we break $x^2 + 5x + 6$ into factors?', 'Can we find where the ball lands using this expression?') indicating readiness for Lessons 3–4; (3) Any green note that describes a quadratic only as 'curved' without mentioning the algebraic structure is noted — teacher will revisit in Lesson 3 opener.
Model Building Phase 📹 **VIDEO:** Video: Area Model Introduction Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their printed grid paper from Lesson 1 where they traced the shot put path. They now write a proposed quadratic expression for the ball's height, using the structure $h = ax^2 + bx + c$, and attempt to substitute the data points ($x = 0$, $h = 1.8$ and $x = 2$, $h = 4.0$ and $x = 4$, $h = 3.8$) to form a system of three conditions. They do not yet solve the system — that comes in later lessons — but they write it down as 'the model we are building'. Below their traced curve on the grid paper, learners annotate: 'This curve is quadratic because it has an x^2 term. The expression has the form $h = ax^2 + bx + c$ where a, b, c are numbers we will find.' Learners also sketch a simple labelled	Say: 'In Lesson 1 you traced the path of the shot put. Now you are going to add the algebra to your drawing — turn your picture into a mathematical model.' Pause for effect, then: 'Write the general form $h = ax^2 + bx + c$ beneath your curve. Now substitute the three data points I give you and write three equations — even if you cannot solve them yet.' Write the three data points on the board: (0, 1.8), (2, 4.0), (4, 3.8). [Allow 7 minutes for independent model-building.] Circulate and narrate what you see: 'I see Njeri has substituted $x = 0$ and got $c = 1.8$ — excellent. Can anyone else confirm that?' [WAIT 10 seconds, cold-call 2 learners.] Say: 'You have just set up a mathematical	Cumulative Model Revision — learners physically annotate the same diagram across all 8 lessons, so the model visibly accumulates algebraic meaning. By substituting one known data point (x=0) immediately and finding $c = 1.8$ without any new instruction, learners experience a small but genuine success that motivates the harder algebra ahead. Naming the diagram a 'mathematical model' frames the unit's purpose: mathematics as a tool for prediction.	Teacher collects or photographs annotated grid papers at the end of the lesson. Observable indicators: (1) Learner has written $h = ax^2 + bx + c$ beneath their traced curve; (2) Learner has correctly substituted (0, 1.8) to obtain $c = 1.8$; (3) Learner has written three substituted equations, even if unsimplified; (4) Learner has labelled both the shot put path and the shamba diagram as 'quadratic'. Learners missing indicators (1) or (2) receive a short written prompt at the top of their returned paper before Lesson 3.

	diagram of the Kericho shamba alongside the shot put path and write: 'Both are modelled by quadratic expressions.' This revised annotated diagram is their cumulative model for the unit — it will be updated every lesson.	model for a real throw at a Nairobi school. In the coming lessons, we will learn the tools to solve for a and b — and then predict exactly where the ball lands.' Close by asking learners to hold up their annotated grid paper: 'This is your model. It will grow every lesson. Keep it safe.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the productive failure in Phase 1 generate genuine cognitive dissonance, or did learners disengage? If disengagement occurred, consider providing a more scaffolded data table with the differences column pre-labelled in Lesson 3. (2) During the jigsaw in Phase 2, which evidence source — the shamba area or the shot put difference table — produced richer discussion? Use the stronger source as the entry point for forming expressions in Lesson 3 (factorisation). (3) Review the DQB orange notes: do learners' new questions cluster around factorisation, graphing, or solving? Adjust the emphasis of Lesson 3 accordingly. (4) Were there learners who could not expand two brackets (e.g., (x)(x+4))? These learners need a brief review of the distributive law at the start of Lesson 3 — note their names. (5) Did the Kenyan contexts (Kericho shamba, Nakuru maize farm, Kisumu mandazi seller) feel authentic and engaging, or did any context cause confusion? Replace any confusing context with a githeri portion-pricing or ugali cooking-pot area scenario in future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners watched the slow-motion shot put video still and worked with a height-distance data table from a throw at a Nairobi school sports day. They also examined a rectangular shamba diagram from Kericho, labelled with side lengths x and (x + 4) metres, and computed first and second differences from the shot put height data.
What did I learn?	Learners discovered that the area of the rectangular shamba produces an expression with an x^2 term when expanded — $x^2 + 4x$ — and that the second differences in the shot put height data are approximately constant, a signature of a quadratic relationship. They identified the standard form $ax^2 + bx + c$ and distinguished it from linear expressions by the presence of the x^2 term (degree 2).
How does this explain the phenomenon?	The curved, parabolic path of the shot put cannot be modelled by a linear expression because the height data does not change by a constant amount per metre of horizontal distance. It requires a quadratic expression — one containing an x^2 term — because the second differences are constant. The same structure ($ax^2 + bx + c$) describes the area of a rectangular shamba when one or both dimensions depend on a variable, confirming that quadratic expressions arise naturally from both motion and area contexts in real Kenyan settings.

LESSON 3 (80 minutes): Factorising Quadratic Expressions — Breaking the Curve's Code	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to factorise quadratic expressions of the form $ax^2 + bx + c$, connecting the factored form to the structure of the shot-put parabola and to real land-area problems in Kericho.
Knowledge	Learners will know: (i) the meaning and structure of a quadratic expression in factored form; (ii) how to factorise expressions of the form $x^2 + bx + c$ by inspection; (iii) how to factorise expressions of the form $ax^2 + bx + c$ ($a \neq 1$) using the product-and-sum method; (iv) how the factors of a quadratic relate to the roots/zeros of its associated curve.
Skills	Learners will be able to: (i) identify factor pairs of the constant term that sum to the coefficient of x; (ii) rewrite and group terms to factorise $ax^2 + bx + c$ systematically; (iii) verify factorisation by expanding back using the identities derived in Lesson 2; (iv) apply factorisation to decode meaningful quantities in a real-life quadratic expression.
Attitudes	Learners will: (i) appreciate that breaking a complex expression into simpler factors is a powerful problem-solving strategy; (ii) show persistence when the first factor-pair attempt does not work; (iii) value collaborative checking and peer correction as part of mathematical practice.
Key Inquiry Question	How do we factorise quadratic expressions?
Purpose in Storyline	In Lesson 2 learners built quadratic expressions from area models and verified the three quadratic identities. They can now expand — but the driving question demands more: to find WHERE the shot put lands (the zeros of the parabola) and to find the DIMENSIONS of a farmer's shamba given its area, learners must reverse the expansion process. Lesson 3 teaches factorisation as that reversal, directly answering the question: 'If we know the curved rule, can we un-pack it to find the values that make height = 0?' Every factorisation technique practised today is motivated by that need.
Safety Notes	No physical hazards. Ensure learners do not share calculators in a way that creates conflict; remind them that peer discussion is encouraged but individual written work must be each learner's own.

B. LESSON OVERVIEW

Lesson 3 sits at the heart of the evidence-gathering sequence. Learners arrive knowing how to form and expand quadratic expressions (Lessons 1–2). Today they discover how to reverse that process — factorisation — through a concrete area-puzzle set in Kericho, verified against the shot-put phenomenon. The lesson opens with a Predict phase where learners try to 'un-expand' a quadratic by inspection, then moves to an Observe phase using a structured area-tile activity and a worked video-still from the sports day. In the Explain phase learners codify the product-and-sum method and apply it to both simple ($a=1$) and general ($a\neq1$) cases. The Driving Question Board is updated to record the new insight that factors reveal zeros. The lesson closes with a cumulative model revision in which learners annotate their parabola sketch with the factored form of the height expression. Throughout, Kenyan contexts (shamba dimensions in Kericho, maize-plot fencing in the Rift Valley, the shot-put arc) keep the mathematics grounded and purposeful.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Adding Decimals by Inserting Place-holding Zeros] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two side-by-side statements on the board: (A) $(x + 3)(x + 5) = x^2 + 8x + 15$ [from Lesson 2 expansion work] and (B) $x^2 + 8x + 15 = ?(?)$. They are asked individually (no discussion yet) to write in their exercise books: 'What two brackets do you think multiply to give $x^2 + 8x + 15$? Write your best guess and explain your reasoning in one sentence.' After 3 minutes of silent thinking, learners share with a shoulder partner for 2 minutes. The teacher then cold-calls 4 learners to share their guesses — without confirming correctness — and records all conjectures on the board under the heading 'Our Predictions'. A second prompt is then placed on the board connected to the phenomenon: 'A classmate writes the height of the shot put above the ground as $h = -x^2 + 6x$ (where x is horizontal distance in metres). She says the ball lands when h = 0. She needs to factorise $-x^2 + 6x$ to find where. What factors do you predict?' Learners write a second prediction silently for 2 minutes.	1. Display both statements (A) and (B) using the board or printed cards. Say exactly: 'In Lesson 2 we learned to EXPAND brackets into a quadratic. Today I want you to try going the other way — from the quadratic BACK to brackets. Do not discuss yet. Write your best guess silently.' (Allow 3 minutes — do not circulate yet; let learners think independently.) 2. After 3 minutes say: 'Now share your guess with your shoulder partner. Listen to their reasoning — you may change your guess if they convince you.' (Allow 2 minutes.) 3. Cold-call exactly 4 learners using name cards or a random selector. Ask each: 'What did you write — and WHY did you choose those brackets?' Record every conjecture on the board verbatim — do not correct. Say: 'I am writing ALL ideas. We will test them during the lesson.' 4. Display the phenomenon prompt about $h = -x^2 + 6x$. Say: 'This expression describes the shot put's height from our sports day video. If we can factorise it, we can find where the ball hits the ground. Write your prediction — silently, 2 minutes.' (WAIT TIME: do not speak for the full 2 minutes.) 5. Tell learners: 'Hold your predictions. We are about to gather evidence to test them.'	Prediction-before-instruction (Elicit Prior Knowledge): By requiring learners to commit a conjecture to paper before any teaching, the teacher activates prior knowledge of multiplication (factor pairs) and creates cognitive dissonance that motivates the Observe phase. Recording all guesses publicly — including incorrect ones — normalises productive struggle and gives the teacher live diagnostic data about misconceptions (e.g., learners who guess (x+8)(x+15) reveal they do not yet understand the product-sum relationship).	Teacher scans written predictions during the 3-minute silent period, noting: (i) learners who correctly identify (x+3)(x+5) — they are ready for the a≠1 challenge; (ii) learners who write (x+8)(x+15) or (x+15)(x+1) — they confuse sum and product; (iii) learners who leave the second prompt blank — they may need a pre-teach on factoring out a common factor. Teacher uses these observations to adjust groupings in Phase 2.
Observe Phase 📹 **VIDEO:** [Adding Decimals by Inserting Place-holding Zeros] Source: CK-12 🔍 Search ARES for similar videos	ACTIVITY A — Kericho Shamba Area Tiles (25 minutes): Learners work in groups of 4. Each group receives a printed 'Shamba Card' showing: 'A rectangular shamba in Kericho has area = $(x^2 + 7x + 12)$ m². The length is (x + 4) m. What is the width?' Groups are given a set of algebra tiles (or a printed grid substitute): one large x^2 tile,	1. Distribute Shamba Cards and tile sets (or printed grid substitutes). Say: 'Your job is to arrange these tiles into a rectangle. The total area is $x^2 + 7x + 12$. One side is already given as (x + 4). Use the tiles to find the other side. You have 15 minutes. Every group member must be able to explain the rectangle.' 2.	Concrete-Representational-Abstract (CRA) with Phenomenon Anchoring: The algebra tile activity provides a CONCRETE manipulation that makes the abstract process of factorisation physically visible — learners see that factorising means finding the side-lengths of a rectangle whose area is known. This directly mirrors	Observable indicators during tile activity: (i) GROUP SUCCESS — group arranges tiles into a correct rectangle with dimensions (x+3) and (x+4); (ii) PARTIAL — group finds one factor by trial but cannot articulate WHY those numbers work; (iii) MISCONCEPTION — group places all 12 unit tiles in a row, showing they do not

📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	seven x-tiles, and twelve unit tiles. Groups must physically arrange the tiles into a rectangle. They record: (i) a diagram of their rectangle, (ii) the two dimensions (factors), (iii) the verification by re-expanding. Groups that finish early receive a second Shamba Card: area = $2x^2 + 7x + 3$. After 15 minutes, a whole-class gallery share takes place — each group holds up their rectangle diagram. ACTIVITY B — Phenomenon Connection (10 minutes): The teacher projects the sports-day video still showing the shot-put arc. The teacher writes on the board: $h = -x^2 + 6x$. Learners are guided to first factor out the common factor: $h = -x(x - 6)$. The teacher asks: 'What does $x = 0$ mean in the context of the throw? What does $x = 6$ mean?' Learners write answers in their books. The teacher connects this to the printed still-frame: 'The ball left the athlete's hand at $x = 0$ and should land at $x = 6$ metres. Does the string measurement on your printed still-frame agree?' Learners check against their Lesson 1 measurements.	Circulate during the activity. At each group, ask one probing question — rotate these three questions: (a) 'How do you know which numbers to pair with the unit tiles?'; (b) 'What multiplication fact links the constant terms of your two factors to the constant in the original expression?'; (c) 'If I change the area to $x^2 + 7x + 10$, would your rectangle change? How?' Give WAIT TIME of 10 seconds after each question before accepting a response. 3. At 15 minutes, stop groups and do a quick gallery share: hold up diagrams. Ask the whole class: 'Are all rectangles the same? Why must they be?' Cold-call 2 groups to explain. 4. For the second Shamba Card ($2x^2 + 7x + 3$), say: 'This one is harder — the coefficient of x^2 is not 1. Work together. You may need to rearrange your tiles more than once.' Do NOT demonstrate the method yet — let groups struggle productively for 5 minutes. 5. Transition to Activity B. Display the video still. Say: 'Let us return to the shot put. I am writing the height expression. Your job: find the two factors. Start by asking — is there anything COMMON to both terms?' (WAIT TIME: 12 seconds.) Cold-call 1 learner. Then say: 'What does each factor tell us about the ball's journey?' Cold-call 3 different learners. 6. Say: 'Check your Lesson 1 string measurement. Does $x = 6$ m match what you measured? Write a one-sentence observation in your book.'	the area-model work from Lesson 2, creating conceptual continuity. The pivot to the shot-put expression in Activity B moves from Concrete (tiles) to Abstract (symbolic factorisation), anchored by the phenomenon. Connecting the factor $x = 6$ to the ball's landing point transforms factorisation from a symbol-manipulation trick into a meaningful prediction tool — exactly what the driving question demands.	understand area as a 2D concept. Teacher notes which groups receive the harder card ($2x^2 + 7x + 3$) and how they approach it — this informs direct instruction focus in Phase 3. For Activity B: listen for learners who correctly state '$x = 6$ means the ball has travelled 6 metres horizontally and is back at ground level' — this is the target conceptual link.
Explain Phase 📹 **VIDEO:** Adding Decimals	WHOLE-CLASS CODIFICATION (15 minutes): The teacher leads the class through a structured	1. Begin by referencing the tile work: 'In your groups you physically found that $x^2 + 7x + 12$	Worked Example with Immediate Verification (Cognitive Apprenticeship): The teacher	During peer practice, teacher collects the shot-put worksheet problem from 5 randomly selected

by Inserting Place-holding Zeros Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	note-making sequence. Using the tile work as a starting point, the teacher introduces the Product-and-Sum method formally. Step 1 — $x^2 + bx + c$ (a=1): 'Find two numbers that MULTIPLY to give c and ADD to give b.' Learners work through three guided examples in their books: (i) $x^2 + 9x + 20$ [Kericho shamba]; (ii) $x^2 - 5x + 6$ [negative b]; (iii) $x^2 + 2x - 15$ [negative c]. After each example, learners swap books and mark each other's work. Step 2 — $ax^2 + bx + c$ ($a \neq 1$): The teacher connects to the harder Shamba Card. Introduces the 'ac method': multiply a × c, find two numbers that multiply to ac and add to b, split the middle term, then factorise by grouping. Learners work two guided examples: (iv) $2x^2 + 7x + 3$ [from the Shamba Card]; (v) $3x^2 - 10x - 8$ [area of a maize plot in the Rift Valley]. PEER PRACTICE (10 minutes): In pairs, learners complete four mixed factorisation problems from a printed worksheet, including one problem that re-contextualises the shot-put height expression: 'The height of a ball thrown at Nairobi School's sports day is $h = -2x^2 + 10x$. Factorise this expression fully and state what the factored form tells you about the throw.'	breaks into (x+3)(x+4). Let us now write a METHOD so we can do this WITHOUT tiles, for any quadratic.' Write the heading 'FACTORISATION — Product and Sum Method' on the board. 2. For each of the three a=1 examples, use the THINK-PAIR-SHOW routine: pose the expression, give 90 seconds silent thinking, 60 seconds pair discussion, then cold-call one learner to come to the board. For EVERY board solution, ask the class: 'Does everyone agree? Who can verify by expanding?' Allow exactly 10 seconds WAIT TIME before taking volunteers. 3. Before introducing the ac method, say: 'Some of you tried $2x^2 + 7x + 3$ with tiles. What made it harder? Tell your partner in 30 seconds.' Cold-call 2 learners. Then say: 'The challenge is that TWO things can change — the number in front of x^2 AND the constant. The ac method handles this.' Write the steps clearly and annotate each step with a spoken explanation. 4. For example (iv), narrate aloud: 'Step 1: a=2, c=3, so ac=6. Step 2: I need two numbers that multiply to 6 and add to 7 — that is 6 and 1. Step 3: Rewrite 7x as 6x + x, giving $2x^2 + 6x + x + 3$. Step 4: Group — 2x(x+3) + 1(x+3). Step 5: Factor out (x+3) — answer is (2x+1)(x+3).' Ask class to verify by expanding. Give 60 seconds WAIT TIME. 5. For peer practice, circulate and target the pairs identified as struggling in Phase 2. For the shot-put problem, ask specifically: 'After factorising, what are the two values of x that make h=0? What do they mean for the athlete?' WAIT 12 seconds before accepting any response. Cold-call	models expert thinking aloud during the ac method, making the invisible reasoning (choice of factor pairs, grouping decision) visible. Immediately requiring learners to verify by expanding closes the expand-factorise loop and reinforces that these are inverse operations. The THINK-PAIR-SHOW routine ensures every learner processes each example before seeing the answer, preventing passive copying. The closing discussion question explicitly re-anchors the technique to the driving question, preventing factorisation from feeling like an isolated algebraic drill.	pairs. Success criteria: (i) correctly factors out $-2x$ from $-2x^2 + 10x$ to get $-2x(x - 5)$; (ii) correctly states x = 0 and x = 5 as the zeros; (iii) interprets x = 5 as the landing distance in metres. Common errors to watch for: (a) forgetting the negative sign when factoring, giving $2x(x-5)$; (b) writing the zeros as x = 2 and x = −10 (substituting coefficients instead of solving); (c) inability to connect zero-factors to the physical landing point.

		3 learners for this question. 6. Close Explain phase by asking: 'How does factorisation connect to our driving question about predicting where the ball lands?' Cold-call 2 learners.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Adding Decimals by Inserting Place-holding Zeros]() Source: CK-12 🔍 [Search ARES for similar videos]() 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)]() Source: CK-12 🔍 [Search ARES for similar readings]()	Each learner receives a sticky note (or writes in the class DQB section of their book). They respond to two prompts: PROMPT 1 (Green sticky — 'What I now understand'): 'Using today's factorisation, write the factored form of $h = -x^2 + 8x$ and explain in one sentence what the two factors tell you about the shot put's journey.' PROMPT 2 (Yellow sticky — 'New question I still have'): 'Write one question that today's lesson has raised for you — something you still want to understand about the parabola or quadratic equations.' Learners place their stickies on the class DQB under the columns 'We now know' and 'We still wonder'. The teacher reads 3 yellow stickies aloud to the class (choosing ones that preview Lesson 4 — forming and solving equations) and says: 'These great questions are exactly what we will investigate in Lesson 4.'	1. Distribute sticky notes (or direct learners to the DQB section). Say: 'The Driving Question asks how mathematics can help us predict where a ball lands. Today we gained a new tool. Use your green note to show how factorisation answers part of that question — be specific, use the expression $h = -x^2 + 8x$.' Give 4 minutes for writing — do NOT accept oral answers; written commitment is required. 2. Say: 'On your yellow note, write one thing you still do not understand, or one question today raised in your mind. There are no wrong questions here.' Give 2 minutes. 3. Collect stickies and place them on the DQB. Read 3 yellow notes aloud without revealing who wrote them. Say: 'These questions show brilliant mathematical thinking. Notice that several of you are asking HOW we actually solve for x — that is exactly where we are going in Lesson 4.' 4. Point to the DQB and say: 'Look at how our board is growing. In Lesson 1 we noticed the curve. In Lesson 2 we built the expression. Today we broke it apart. Each lesson we get closer to a complete answer.'	Driving Question Board as Cumulative Knowledge Tracker: The DQB makes the progression of understanding visible and public over the eight-lesson sequence. Green stickies externalise consolidation; yellow stickies reveal genuine curiosity and diagnostically inform lesson sequencing. Publicly reading yellow stickies — without attribution — validates intellectual struggle and signals that unanswered questions are the engine of further inquiry, not evidence of failure.	Teacher reads all green stickies after the lesson. Success indicator: learner writes the factored form $-x(x - 8)$ and connects the two zeros ($x=0$ and $x=8$) to the ball's launch point and landing point. Partial success: learner factorises correctly but does not interpret the factors in context. Concern: learner cannot factorise or interprets the expression as describing a linear relationship. Yellow stickies are coded by theme: 'How do we solve for x?' (previews Lesson 4), 'What if the ball is thrown upward at an angle?' (previews real-life quadratic equations), 'Can a quadratic have no zeros?' (previews the discriminant concept) — these guide teacher planning.
Model Building Phase 📹 **VIDEO:** [Adding Decimals by Inserting Place-holding]()	Learners retrieve their cumulative parabola sketch begun in Lesson 1 (a hand-drawn arc of the shot-put path on a printed grid, annotated in Lesson 2 with the expanded quadratic expression for height). Today they make TWO additions	1. Say: 'Take out your parabola sketch from Lesson 1, with the Lesson 2 annotation. You are about to make it more powerful.' Give learners 2 minutes to locate and review their previous annotations. 2. Say: 'First addition:	Cumulative Model Revision (Progressive Modelling): Each lesson adds a layer of mathematical meaning to the same physical sketch, making the growth of understanding visible to the learner themselves. By	Teacher collects 6 models (two from each ability tier identified in Phase 2) for detailed review after the lesson. Criteria: (i) Both zeros correctly identified and labelled on the sketch — indicates understanding that factors yield

Zeros Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	to the model: (i) Below the expanded form, they write the factored form of the height expression and circle the two x-values that make h = 0, labelling them on the horizontal axis of their sketch as 'Launch point' and 'Landing point'. (ii) In the margin, they add a 'Kericho Connection' box: they write the factored form of $x^2 + 7x + 12 = (x+3)(x+4)$ and sketch the corresponding shamba rectangle with labelled dimensions, noting: 'Factorising a quadratic = finding the side-lengths of the rectangle with that area.' Learners then write a one-paragraph 'Model Update Statement' below their sketch: 'My model of the shot-put parabola has changed because...' They should reference both the factored form AND the physical meaning of the zeros.	write the factored form of your height expression underneath the expanded form. Then find the two values of x that make h = 0. Mark them on your sketch's horizontal axis with a small circle and label them.' Give 5 minutes. Circulate and check that learners are marking BOTH x=0 (launch) and the positive zero (landing). If a learner marks only one zero, ask: 'The ball starts at a height of zero — where on your sketch is that moment?' WAIT 10 seconds. 3. Say: 'Second addition — the Kericho Connection. In the margin, sketch the shamba rectangle with dimensions from today's tile activity and label the sides. Then write the sentence in the template.' Give 4 minutes. 4. Say: 'Finally, write your Model Update Statement. Start with: My model of the shot-put parabola has changed because... You must mention the word FACTORS and the word ZEROS.' Give 4 minutes for writing. 5. Cold-call 2 learners to read their Model Update Statements aloud. Ask the class after each: 'Does their statement capture how factorisation connects to the parabola? What would you add?' (WAIT TIME: 10 seconds before responses.) 6. Close by saying: 'Your model now shows the curve, the expression, AND where the curve crosses the ground. In Lesson 4 we will set up a full equation and solve it — so we can predict the landing point for ANY throw, even one we have never seen.'	requiring both the factored form AND the zeros to appear on the same diagram, the model forces the learner to connect symbolic manipulation to geometric meaning. The 'Kericho Connection' box prevents factorisation from being isolated in the algebra strand — it explicitly links the technique to the area-model context from Lesson 2, reinforcing coherence across the evidence-gathering sequence. The Model Update Statement is a metacognitive writing task that consolidates learning and prepares learners for the assessment portfolio.	zeros; (ii) Factored form written correctly and matches the expanded form from Lesson 2 — indicates factorisation accuracy; (iii) Shamba rectangle drawn with correct labelled dimensions — indicates transfer of factorisation to a new context; (iv) Model Update Statement uses both required terms (factors, zeros) in a mathematically accurate sentence — indicates conceptual integration. Teacher returns annotated models at the start of Lesson 4 with one written question to prompt further thinking.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) FACTORISATION FLUENCY — What proportion of learners successfully factorised a=1 expressions independently? What proportion attempted the a≠1 case? Were there common error patterns (e.g., sign errors when c is negative) that need re-addressing at the start of Lesson 4? (2) PHENOMENON CONNECTION — When learners updated their parabola models, did they spontaneously connect the zeros of the factored expression to the ball's landing, or did this require heavy scaffolding? If scaffolding was needed, consider adding an explicit 'zero-product property' mini-lesson at the opening of Lesson 4. (3) TILE ACTIVITY MANAGEMENT — Did the algebra tile / grid activity run efficiently in groups of 4? Were any groups off-task during the harder $2x^2+7x+3$ card? Consider whether reducing to pairs for the observation phase would increase individual accountability. (4) DQB QUALITY — Read the yellow stickies carefully. Do learner questions cluster around 'how to solve' (well-positioned for Lesson 4) or 'what is a quadratic' (suggests conceptual gaps from Lessons 1–2 that need addressing)? (5) KENYAN CONTEXT — Did the Kericho shamba context feel authentic to learners, or did it feel forced? Note any learner comments that suggest richer local contexts to use in future lessons (e.g., learners who mention their own family's farm plots or local athletics events).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners arranged algebra tiles into rectangles to find the factor-pairs of quadratic expressions, and connected the factored form of $h = -x^2 + 6x$ to the shot put's launch and landing points on the sports-day video still.
What did I learn?	Factorising a quadratic expression (both the a=1 and a≠1 cases using the product-sum and ac methods) reverses the expansion process; the two factors of a height expression reveal the exact horizontal distances where the ball is at ground level (height = 0).
How does this explain the phenomenon?	Learners annotated their cumulative parabola model with the factored form of the height expression and labelled the two zeros on the horizontal axis, and wrote a Model Update Statement explaining how factorisation connects to predicting the ball's landing point.

LESSON 4 (40 minutes): Squaring the Shamba — Deriving Quadratic Identities Through Area	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive the three standard quadratic identities — $(a+b)^2$, $(a-b)^2$, and $(a+b)(a-b)$ — geometrically by cutting and rearranging squared graph-paper pieces representing shamba plots, then connect each identity to the arc of the shot put.
Knowledge	– State and write the three quadratic identities: $(a+b)^2 = a^2 + 2ab + b^2$; $(a-b)^2 = a^2 - 2ab + b^2$; $(a+b)(a-b) = a^2 - b^2$ – Explain how each identity is derived from the concept of area on squared graph paper – Distinguish between the three identities and identify which applies in a given expression
Skills	– Cut, label, and rearrange squared graph-paper pieces to derive each identity geometrically – Apply the three identities to expand and simplify quadratic expressions without multiplying term by term – Connect the identity $(a+b)(a-b) = a^2 - b^2$ to the factorised form of a quadratic expression
Attitudes	– Appreciate that algebraic shortcuts (identities) arise from geometric reasoning, not arbitrary rules – Show persistence and precision when handling cut-out pieces and verifying algebraic results
Key Inquiry Question	How do we form quadratic expressions — and can we find faster, pattern-based shortcuts for special products?
Purpose in Storyline	Having expanded general quadratics in Lesson 2 and factorised them in Lesson 3, learners now discover that certain quadratic expressions have elegant geometric structures — perfect squares and difference of two squares — that produce fast-track identities. These identities will later be used as a factorisation shortcut and will appear directly in the algebraic model of the shot put's parabolic path.
Safety Notes	Learners use scissors to cut graph paper. Remind learners to handle scissors safely — cut away from the body and pass scissors handle-first. No other specific hazards.

B. LESSON OVERVIEW

In this lesson, learners move from the general factorisation work of Lesson 3 into the special, elegant world of quadratic identities. The anchoring context — a farmer in Kericho enlarging a square shamba — provides a concrete, area-based entry point for each identity. Learners physically cut squared graph paper into labelled regions (a^2, $2ab$, b^2 pieces) and rearrange them to prove that $(a+b)^2 = a^2 + 2ab + b^2$ is not just an algebraic rule but a geometric truth. The same physical manipulation is then used to derive $(a-b)^2$ and, through the idea of "removing a square corner from a square shamba," the difference of two squares identity $(a+b)(a-b) = a^2 - b^2$.

The lesson is structured so that the geometry leads the algebra: learners see the area first, write the expression second, and only then recognise the identity as a shortcut. This sequence builds conceptual understanding and prevents the common error of writing $(a+b)^2 = a^2 + b^2$ — an error that arises precisely when learners skip the geometric step. Each identity is immediately connected back to the shot put phenomenon: the teacher explicitly asks learners whether they have seen any of these patterns hiding inside the quadratic height expressions from previous lessons.

By the end of the lesson, learners can state all three identities, explain why each is true using an area diagram, and add two high-quality factual and conceptual questions to the Driving Question Board. The cumulative model is updated to show that some quadratic expressions have special structure that can be read directly from their form, without full factorisation — a powerful new tool for decoding the shot put's parabola.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a small strip of squared graph paper. The teacher draws a square of side (a+b) on the board — labelled as a Kericho tea farmer's square shamba of side (3+2) = 5 metres — and asks: 'If I square the total side length, I get 25 m². But what if I split the side into two parts — 3 m of maize and 2 m of beans — and square each part separately? Will I get the same answer?' Learners write their prediction (Yes / No / Sometimes) and a brief reason in their exercise books before any discussion.	Display the diagram on the board with clear labels: total side = (a + b), inner square side = a, border strip width = b. Say exactly: 'I want you to predict — not calculate yet, just predict — whether $(a + b)^2$ is the same as $a^2 + b^2$. Write your answer and your reason. You have 60 seconds. Go.' [WAIT TIME: 60 seconds, no prompts.] After writing, cold-call 3 learners: one who wrote YES, one who wrote NO, one who wrote SOMETIMES (scan the room quickly). Ask each: 'Tell us your reasoning.' Do not confirm or correct — simply record the three positions on the board as 'Class Predictions' and say: 'We are going to settle this argument using scissors and graph paper. The shamba will tell us who is right.'	Predict–Observe–Explain (POE) Launch: By requiring a written prediction before any instruction, this strategy activates prior knowledge (area of squares and rectangles from Junior School), surfaces the very common misconception $(a+b)^2 = a^2 + b^2$, and creates cognitive dissonance that motivates the observation phase. Recording three competing predictions publicly raises the stakes of the investigation.	Teacher scans written predictions as learners write. Key indicator: learners who write $(a+b)^2 = a^2 + b^2$ (the target misconception) — note these learners by name for focused follow-up during the Observe phase. A correct prediction with geometric reasoning ('the area includes the middle rectangle parts') signals readiness to move faster.
Observe Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In pairs, learners receive pre-printed squared graph-paper templates (or draw their own on 1 cm² paper). They cut out: one large square of side a (use a = 4 cm), one small square of side b (use b = 2 cm), and two identical rectangles each of dimensions a × b. Task 1: Arrange all four pieces to form one large square of side (a+b). Learners label each piece with its area expression (a^2, b^2, ab, ab) and record the total area as $a^2 + 2ab + b^2$. Task 2: Teacher then poses the shamba-reduction scenario: 'The farmer decides to reduce the shamba by b metres on each side — the new shamba has side (a−b). What is its area?' Learners now fold or redraw to model $(a-b)^2$, again labelling	Distribute scissors and graph paper. Say: 'Cut carefully. You are building mathematical proof with your hands.' Circulate during Task 1. After 4 minutes, pause the class and cold-call one pair to hold up their arrangement: 'Show us your square. What are the four pieces? What is the total area?' [WAIT TIME: 10 seconds after each question.] For Task 2, say: 'Now the rains failed in Kericho and the farmer must shrink the shamba. Fold away a strip of width b from two sides. What shape do you have? What is its area? Write the identity.' Circulate — look for pairs who are subtracting only one strip. Prompt: 'Did you subtract from both sides? How many strips of b did you remove?' For Task 3, hold	Geometric Area Modelling (Concrete–Pictorial–Abstract progression): Learners move through three modes — physical cut-outs (concrete), labelled diagrams (pictorial), and algebraic identity statements (abstract). This CPA sequence is the KICD-recommended approach for deriving identities 'from the concept of area' and directly mirrors the assessment indicator: 'The learner correctly derives quadratic identities from the concept of area.'	Teacher checks that each pair's structured table has correct algebraic forms for all three identities. Specific indicators: (1) Does the $(a+b)^2$ row show '+ 2ab' and not '+ ab'? (2) Does the $(a-b)^2$ row show '− 2ab'? (3) Does the difference-of-two-squares row show both the factored form (a+b)(a−b) AND the expanded form $a^2 - b^2$? Pairs with errors in the sign of the middle term receive a targeted question: 'How many ab rectangles did you count in your cut-out?'

	pieces. Task 3: The teacher introduces the 'L-shaped shamba': a square of side a with a square corner of side b removed. Learners measure and write the area of the L-shape two ways — as $(a-b)(a+b)$ by factoring the L-shape into two rectangles — and as $a^2 - b^2$ by subtraction. All findings are recorded in a structured table: Identity | Geometric Meaning | Algebraic Form.	up the L-shaped cut-out and say: 'This is a shamba with a corner sold off. I can calculate its area two ways. Work in pairs — find both ways and see if they match.' [WAIT TIME: 2 minutes working time.] Cold-call 2 pairs to share. Confirm with: 'Both ways give the same area — that is not a coincidence. That IS the identity.'		
Explain Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners first apply each identity to numerical checks: they verify $(5+2)^2 = 49$ using the identity ($a^2 + 2ab + b^2 = 25 + 20 + 4 = 49$ ✓) and $(7-3)^2 = 16$ using ($49 - 42 + 9 = 16$ ✓) and $(8+3)(8-3) = 55$ using ($64 - 9 = 55$ ✓). Then they apply the identities algebraically to three expressions drawn directly from the shot put and shamba storyline: (i) Expand $(x+5)^2$ — relating it to a shamba whose side is extended by 5 m; (ii) Expand $(x-3)^2$ — relating it to the shot put height expression at a shifted launch point; (iii) Factorise $x^2 - 16$ using the difference of two squares — which the teacher explicitly connects to: 'This is the kind of expression we might get if we rearrange the shot put height equation.' Learners write the full working in their books, stating which identity they used and why.	Write the three worked examples on the board one at a time — do not show all three simultaneously. For each, say: 'Which identity applies here? Tell your partner first.' [WAIT TIME: 15 seconds.] Then cold-call one learner to identify the identity and one learner to complete the first step. For the difference of two squares, explicitly revisit the phenomenon: 'Cast your mind back to Lesson 3. When we factorised the shot put height expression and set it equal to zero, we got two values of x — the landing points. Now look at $x^2 - 16$. Can you factorise this using your new identity without using the ac method?' [WAIT TIME: 20 seconds.] Cold-call 2 learners. After correct responses, say: 'This is the power of an identity — it is a shortcut that saves time and reveals structure. A mathematician in Nairobi and a farmer in Kericho both use this same pattern.' Distribute a short written task: three expressions to expand/factorise independently (5 minutes, individual silent work). Circulate and mark three books per minute.	Storyline Anchoring with Graduated Application: By moving from numerical verification → shamba-context expansion → shot put factorisation, this strategy ensures the identities are not learned as isolated rules but are immediately embedded in both the supporting phenomenon (shamba) and the anchoring phenomenon (shot put arc). Each step increases abstraction while maintaining the storyline thread, building what KICD describes as 'ability to form quadratic expressions' through purposeful application.	Collect the 5-minute individual written task. Look for: (1) Correct identification of which identity to use before expanding — learners who launch straight into term-by-term multiplication have not internalised the identity. (2) Correct sign on the middle term of $(x-3)^2$: the result must be $x^2 - 6x + 9$, not $x^2 + 6x + 9$. (3) Correct factorisation of $x^2 - 16$ as $(x+4)(x-4)$. Any learner writing $(x-4)^2$ has confused difference of two squares with a perfect square — flag for re-teaching in the next lesson.
Driving	Learners look at the class Driving	Give learners 3 minutes to write	Driving Question Board —	Teacher reads all posted factual

Question Board (DQB) Creation 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Question Board and revisit the unit driving question: 'How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems?' Each learner writes two questions on sticky notes (or index cards): one FACTUAL question (something they now know the answer to, e.g., 'What is the formula for $(a-b)^2$?') and one CONCEPTUAL question (something they are still wondering, e.g., 'Can the difference-of-two-squares identity help us find where the shot put lands more quickly than the ac method?'). Learners post both notes on the DQB under the columns 'What We Now Know' and 'What We Are Still Wondering.'	their two questions. Say: 'Your factual question should be something you could now answer yourself. Your conceptual question should be something that genuinely puzzles you about the shot put or the shamba — something today's lesson raised but did not yet fully answer.' After posting, the teacher reads 3–4 conceptual questions aloud, selecting ones that preview upcoming lessons (e.g., questions about solving equations or finding the landing point). Say: 'These questions are exactly where we are headed. By Lesson 6, you will be able to answer every one of these.' Point to the DQB trajectory and say: 'Notice how our board has grown since Lesson 1. We started with one question — what kind of curve is this? — and now we have a whole web of connected questions. That is what real mathematical thinking looks like.' Do not answer the conceptual questions — explicitly defer them.	Differentiated Posting (Factual vs. Conceptual): Requiring one question of each type prevents the DQB from becoming a list of things already known. The conceptual questions generate productive intellectual tension — learners articulate what they do not yet understand, which is the precondition for deep learning. Publicly displaying and reading selected questions validates learner curiosity and creates a social contract: the class will return to these questions.	questions before the next lesson. Indicator of understanding: factual questions should correctly state an identity (e.g., '$(a+b)(a-b) = a^2 - b^2$'). Factual questions that contain errors (e.g., '$(a+b)^2 = a^2 + b^2$') reveal persistent misconceptions and must be addressed explicitly at the start of Lesson 5. Conceptual questions that mention 'landing point,' 'solving,' or 'equals zero' indicate learners are already connecting identities to equation-solving — these learners may be extended in Lesson 5.
Model Building Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative algebraic model of the shot put's path. In Lesson 1 they described the curve qualitatively; in Lesson 2 they wrote $h = -x^2 + 6x$ (or similar); in Lesson 3 they factorised it. Now they annotate their model diagram with a new layer: they examine whether any part of the height expression can be rewritten using an identity. Specifically, they complete the square mentally: noting that $-x^2 + 6x = -(x^2 - 6x) = -[(x-3)^2 - 9] = -(x-3)^2 + 9$, and mark on their parabola sketch the point $x = 3$ as the axis of symmetry and $h = 9$ as the maximum height. Learners label this on their diagram with the annotation:	Say: 'We have been building a model of the shot put's arc since Lesson 1. Every lesson adds a new layer. Today's layer is: can any of our three identities help us read the model more deeply?' Write on the board: $h = -x^2 + 6x$. Say: 'Factor out the negative sign first. What do you get inside the bracket?' [WAIT TIME: 15 seconds.] Cold-call 1 learner. Then say: 'Now look at $x^2 - 6x$. Does this remind you of any of the three identities from today?' [WAIT TIME: 20 seconds.] Cold-call 2 learners. Guide them: 'To make it a perfect square, what number am I missing? Think about $(x - 3)^2$.' Allow 1 minute of paired	Cumulative Model Revision (Layer-by-Layer Annotation): Each lesson adds a new analytical layer to the same shot put model. This strategy builds a coherent, growing representation of the phenomenon rather than isolated lesson tasks. The visible history of the model — different colours per lesson, growing annotations — communicates to learners that science and mathematics are cumulative, iterative processes. The discovery of the turning point via the perfect square identity creates a 'wow moment' that motivates completing the square in later lessons.	Teacher circulates and checks that model diagrams include: (1) the correct factored-out form $-(x^2 - 6x)$; (2) the identity annotation $(x-3)^2$ with the correct expansion check; (3) the turning point labelled at (3, 9) on the parabola sketch. Learners who cannot connect $-(x-3)^2 + 9$ to the turning point are not yet expected to fully understand completing the square — this is a preview. The key indicator at this stage is simply that they recognise the perfect square identity in the expression and write it correctly.

	'Identity used: $(x-3)^2$ — perfect square form reveals the turning point.' This is an introduction to completing the square, which will be formalised in a later lesson — here it is treated as a discovery.	discussion. Then reveal the completed annotation. Say: 'The highest point of the ball — 9 metres above the ground at $x = 3$ metres — was hiding inside the identity all along. This is why mathematicians love identities: they reveal structure that is invisible in the original form.' Learners update their model in their exercise books — drawing, labelling, and annotating in a different colour ink to show Lesson 4's contribution.		

D. TEACHER REFLECTION
1. During the Predict Phase, how many learners predicted $(a+b)^2 = a^2 + b^2$ (the target misconception)? Did the cut-out activity in the Observe Phase visibly shift those learners' understanding — and how do I know? 2. Were all three identities derived by learners through the geometry, or did I find myself telling them the identity before they had discovered it through the cut-outs? What would I adjust to give learners more discovery time? 3. In the Explain Phase, could learners correctly identify which of the three identities to apply before expanding — or did they default to term-by-term multiplication? What does this tell me about the depth of their conceptual understanding versus procedural fluency? 4. What patterns did I notice in the DQB conceptual questions? Did any learners already connect the difference-of-two-squares to solving equations (setting equal to zero)? How can I use these questions to differentiate in Lesson 5? 5. In the Model Building Phase, did the 'completing the square preview' create productive curiosity or confusion? Should I revisit this opening before Lesson 7, or can I leave it as a planted seed?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners physically cut and rearranged squared graph-paper pieces representing a Kericho farmer's square shamba, observing that a square of side $(a+b)$ always decomposes into exactly four regions — a^2, ab, ab, and b^2 — no matter what values a and b take. They also observed that removing a square corner from a larger square produces an L-shaped area equal to $(a+b)(a-b)$, confirming that $a^2 - b^2$ factors into two binomials.
What did I learn?	The three standard quadratic identities — $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$, and $(a+b)(a-b) = a^2 - b^2$ — are not arbitrary rules but geometric truths rooted in area. The middle term '2ab' in the perfect square identities represents the two rectangle strips that are always present when a square is partitioned, and its sign depends on whether the side is being extended or reduced. The difference of two squares identity provides a rapid factorisation shortcut for any expression of the form $a^2 - b^2$.
How does this explain the phenomenon?	The shot put's height expression $-x^2 + 6x$ can be rewritten using the perfect square identity as $-(x-3)^2 + 9$, revealing that the ball reaches a maximum height of 9 metres at a horizontal distance of 3 metres — the turning point of the parabola. This shows that the identities derived from the shamba's area are not separate from the shot put phenomenon: they are tools for reading the hidden structure of the quadratic expression that describes the ball's arc, and they point directly toward the concept of solving quadratic

	equations (finding where h = 0) in the next lessons.

LESSON 5 (80 minutes): Factorising Quadratic Expressions — Unlocking the Hidden Structure of the Curve	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to factorise quadratic expressions of the form $ax^2 + bx + c$ and connect this algebraic skill to identifying key features of the shot put parabola — particularly where the ball's height equals zero (ground level).
Knowledge	Learners will correctly factorise quadratic expressions including expressions of the form $x^2 + bx + c$ and $ax^2 + bx + c$ (where $a > 1$) using the product-sum method and grouping, and will identify factors as related to roots/zeros of the expression.
Skills	Learners will systematically factorise quadratic expressions, verify factorisation by re-expanding, and apply factorisation to identify where a quadratic expression equals zero in a real-life context.
Attitudes	Learners will appreciate that the factorised form of a quadratic expression reveals hidden structure — specifically the points where a parabolic path meets the ground — and will demonstrate persistence when working through multi-step factorisation problems.
Key Inquiry Question	How do we factorise quadratic expressions?
Purpose in Storyline	In Lessons 1–4, learners formed quadratic expressions, expanded brackets, and recognised the standard form $ax^2 + bx + c$ as the algebraic fingerprint of the shot put path. Now in Lesson 5, learners ask the reverse question: if we are given the expanded expression, can we 'un-expand' it back into factors? This matters for the phenomenon because factorising $h(x) = -x^2 + 6x$ (a simplified height model) immediately reveals the two x-values where height = 0 — exactly where the ball leaves the athlete's hand and where it lands. Factorisation is therefore not an abstract drill; it is the tool that tells us the landing point of the throw before the ball even hits the ground.
Safety Notes	No physical hazards. Ensure learners do not share writing materials in a way that causes conflict; each pair should have their own worksheet. If using printed graph strips from Lesson 4, handle paper edges carefully.

B. LESSON OVERVIEW
Lesson 5 is the pivotal evidence-gathering lesson in which learners discover that quadratic expressions can be 'un-expanded' — factorised — and that the factors directly reveal where the parabolic path of the shot put crosses ground level. Learners begin by predicting which pairs of brackets could produce a given expanded expression, then investigate the product-sum pattern through a structured group activity using Kenyan number contexts (maize-bag dimensions, shamba boundaries). They connect each factorisation back to the shot put height model introduced in Lessons 1–4, building toward solving quadratic equations in Lessons 6–8. The lesson uses the NGSS practices of Analysing and Interpreting Data (Practice 4) and Using Mathematics and Computational Thinking (Practice 5). Formative checkpoints are embedded at every phase. The Driving Question Board is updated with the new insight: 'The factors tell us where the ball lands.'

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Subtracting by Grouping]	Each learner receives a card showing the expanded quadratic expression $h(x) = x^2 + 5x + 6$. The teacher projects the shot put still-	1. Project the shot put still-frame and say verbatim: 'On Sports Day in Nairobi, we watched the ball trace a perfect arch. We wrote its	Activation of Prior Knowledge via Prediction: By asking learners to reverse-engineer the brackets before instruction, the teacher	Observable indicator: Can the learner write any plausible pair of brackets (even if incorrect)? A learner who writes '(x + ?)(x + ?)' is

(Numbers up to 10) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	frame image from Lesson 1 and asks: 'We know this expression describes a curved path. I want you to PREDICT — working alone for 3 minutes — what two bracket expressions, when multiplied together, might give us $x^2 + 5x + 6$. Write your prediction in your notebook. You do not need to be correct; you need to think carefully.' Learners write their bracket guesses privately. After 3 minutes, the teacher asks learners to share with an elbow partner, then cold-calls 4 learners to share predictions with the class (including at least one incorrect prediction). Teacher scribes all predictions on the board without evaluating them, saying: 'We will return to these at the end of Phase 3 and see who was closest.'	path as $x^2 + 5x + 6$. Today we ask: can we reverse that — can we take the expanded form and find the hidden brackets inside it?' 2. Distribute prediction cards. Allow 3 minutes silent thinking — enforce silence and wait. 3. Say: 'Turn to your elbow partner. Share your prediction. 30 seconds.' 4. Cold-call exactly 4 learners using a random name strategy (e.g., picking folded name-strips from a tin). Write all predictions verbatim on the board. 5. Say: 'None of these predictions are wrong yet — they are all hypotheses. Keep them in mind.' Allow 10 seconds WAIT TIME after each cold-call before scribing.	surfaces existing mental models about multiplication and factors. This creates cognitive dissonance when predictions differ — motivating the investigation in Phase 2. Connecting to the phenomenon ensures the algebra feels purposeful, not procedural.	showing structural awareness of binomial factors. A learner who writes a single bracket or a trinomial is still thinking additively — flag these learners for targeted support in Phase 2. Teacher circulates and notes at least 3 specific learner responses during the prediction phase.
Observe Phase 📹 **VIDEO:** Subtracting by Grouping (Numbers up to 10) Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of 4. Each group receives a FACTORISATION INVESTIGATION WORKSHEET with three sections. Section A: Product-Sum Table — groups complete a table where they find two numbers that multiply to give 'c' and add to give 'b' for six different expressions (e.g., $x^2 + 7x + 12$: find two numbers that multiply to 12 and add to 7). Kenyan context clues are embedded: 'A farmer in Kericho is dividing a rectangular tea shamba of area $(x^2 + 7x + 12)$ m² into two strips. What could the length and width expressions be?' Section B: Grouping Method — groups work through two guided examples of factorising $ax^2 + bx + c$ where $a > 1$ (e.g., $2x^2 + 7x + 3$) using the split-the-middle-term and grouping method, following annotated step cards. Section C: Verification —	1. Say: 'You are now detectives. Your job is to find the hidden factors inside these expressions. Use the Product-Sum table in Section A — find two numbers that multiply to the last term and add to the middle term.' 2. Allow 10 minutes for Section A. Circulate — do NOT give answers; instead ask: 'What pairs of numbers multiply to give 12? List them all. Now which pair also adds to 7?' (WAIT TIME: 12 seconds after each probe question.) 3. After 10 minutes, pause the class: 'Hold your pens. Cold-call: Amina, what two numbers did your group find for $x^2 + 7x + 12$? [WAIT 10 seconds]. Brian, do you agree? Why?' 4. Model Section B example $(2x^2 + 7x + 3)$ live on the board using the split-middle-term method. Narrate every step aloud. Say: 'We split 7x into 6x + x because $6 \times 1 = 2 \times 3 = 6$ — the product of a and c.' 5.	Structured Inquiry with Scaffolded Worksheet (NGSS Practice 4 — Analysing and Interpreting Data): The product-sum table provides a concrete data-collection experience before abstraction. Learners are not told the rule; they discover the product-sum relationship by completing the table across six cases. The Kericho shamba context grounds each expression in a physical rectangle, reinforcing that factors represent dimensions — exactly as the area-model derived quadratic identities in Lesson 2. Verification by re-expansion (Section C) closes the loop, making factorisation falsifiable: if re-expansion fails, the factorisation is wrong.	Observable indicators: (a) In Section A, can the learner systematically list factor-pairs rather than guessing randomly? (b) In Section B, can the learner correctly split the middle term using the product ac? (c) In Section C, does the learner's re-expansion match the original — and do they spot their own errors? Teacher uses a 3-colour sticky-dot system on circulating: green dot on worksheet = correct product-sum found; yellow = partial (one error); red = needs re-teaching on factor pairs. After the phase, count dots to identify which groups need targeted support before Phase 3.

	groups re-expand their factorised answers using the FOIL method learned in Lesson 4, confirming they recover the original expression. Each group records findings on a large manila sheet for gallery sharing.	Allow 12 minutes for groups to complete Sections B and C. 6. Cold-call 2 groups to present one factorisation each from Section B. Ask: 'How does your verification in Section C prove your factorisation is correct?'		
Explain Phase 📹 **VIDEO:** [Subtracting by Grouping (Numbers up to 10)] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher brings the class back to the anchoring phenomenon. The simplified shot put height model $h(x) = -x^2 + 6x$ is projected (introduced in Lesson 3). The teacher says: 'This expression gives the height of the shot put at horizontal distance x metres. We need to know: at what value of x does the height equal zero? That is, where does the ball land?' Learners are guided through factorising $h(x) = -x^2 + 6x$ step by step: first factoring out $-x$ to get $-x(x - 6)$, then identifying the two values of x that make $h(x) = 0$ ($x = 0$ and $x = 6$). The teacher connects each step back to the real event: '$x = 0$ is where the athlete releases the ball at ground level; $x = 6$ means the ball travels 6 metres horizontally before landing.' Learners then independently factorise three new quadratic expressions on a mini-whiteboard or in their notebooks: (1) $x^2 + 8x + 15$, (2) $x^2 - 5x + 6$, (3) $3x^2 + 10x + 3$. After each, a random learner is cold-called to explain their method aloud. Finally, learners return to their Phase 1 predictions on the board. The teacher cold-calls 2 learners to evaluate whether their prediction was correct and explain why.	1. Project $h(x) = -x^2 + 6x$ over the shot put still-frame. Say verbatim: 'This is our athlete's throw from Sports Day. The expression $-x^2 + 6x$ tells us the height at every point. But where does the ball land? Factorisation will answer that — today, right now.' 2. Walk through factorisation of $-x^2 + 6x$ on the board, narrating every step. Say: 'I first look for a common factor in every term. Every term has x, and every term has a negative — so I factor out $-x$.' Write: $-x^2 + 6x = -x(x - 6)$. 3. Say: 'Now I ask: what values of x make this zero?' WAIT TIME: 15 seconds. Cold-call: 'Juma, if $-x = 0$, what is x?' WAIT 10 seconds. 'And if $(x - 6) = 0$, what is x?' 4. Connect to phenomenon: 'So $x = 0$ is the release point. $x = 6$ is the landing point. Our athlete's throw travels exactly 6 metres. Factorisation just told us that — before we even measured it.' 5. Set the three independent practice problems. Circulate for 8 minutes. After each problem, cold-call 1 learner: 'Wanjiru, show us your factors for number 2. Explain why those numbers work.' WAIT 12 seconds before accepting the response. 6. Return to the Phase 1 prediction board. Ask: 'Who predicted $(x+2)(x+3)$ for $x^2 + 5x + 6$? Let's check: expand it. Yes — correct! What made that prediction work?' Cold-call 2 learners.	Phenomenon Reconnection with Worked Example (NGSS Practice 5 — Using Mathematics and Computational Thinking): The teacher explicitly solves a real sub-problem of the driving question (where does the ball land?) using factorisation. This demonstrates that the algebraic technique has direct explanatory power over the physical event. The return to Phase 1 predictions closes the predict-observe-explain loop, reinforcing metacognitive awareness: learners see which of their initial hypotheses were structurally sound and why.	Observable indicators: (a) Can the learner correctly factorise all three independent practice problems without prompting? (b) Can the learner articulate verbally why the two numbers in the factors must satisfy the product-sum condition? (c) Can the learner connect $x = 0$ and $x = 6$ to the physical meaning of launch and landing? Teacher collects mini-whiteboards (or scans notebooks) after Phase 3 to identify persistent errors — particularly sign errors in expressions with negative b (e.g., $x^2 - 5x + 6$).

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Subtracting by Grouping (Numbers up to 10)] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (different colours). On the ORANGE note, they write one thing they now understand about the parabolic path that they did not understand before today. On the GREEN note, they write one new question this lesson has raised. Learners post their notes on the class DQB, which is divided into three columns: WHAT WE KNOW | WHAT WE WONDER | HOW THIS CONNECTS TO THE THROW. The teacher reads 4–5 orange notes aloud and, using a marker, draws an arrow on the DQB from 'factorisation' to a pre-posted card that reads 'WHERE DOES THE BALL LAND?' and writes '→ x = 0 and x = 6 (from factorisation!)' in red. The teacher then reads 3 green wonder-notes aloud and circles any that point forward to Lesson 6 (solving quadratic equations). The teacher says: 'These green questions — especially this one about what happens when a ≠ 1 — are exactly what we will investigate next lesson when we form and solve full quadratic equations.'	1. Distribute sticky notes. Say: 'Two minutes — silent. Orange: what do you now understand about the shot put path? Green: what new question has today raised for you?' Enforce silence. 2. Direct learners to post notes on the DQB. Circulate and read notes as they are posted. 3. Read 4–5 orange notes aloud, paraphrasing: 'Achieng writes: I now know that the factors tell us where the ball hits the ground. Exactly — that is our biggest insight today.' 4. Draw the red arrow on the DQB connecting 'factorisation' to 'where does the ball land?' 5. Read 3 green wonder-notes. Say: 'Keep these questions — they are your compass for the rest of this unit.' 6. Cold-call 1 learner: 'Otieno, in one sentence — what does factorising a quadratic expression tell us about the curved path of the throw?' WAIT 15 seconds.	Driving Question Board as a Living Knowledge Artefact: The DQB externalises the class's growing understanding and makes visible the gap between current knowledge and the full driving question. Colour-coding (orange = knowledge, green = wonder) helps learners distinguish claims from questions — a core scientific practice. The teacher's red arrow makes the causal link between today's mathematical tool (factorisation) and the physical question (landing distance) permanently visible in the classroom.	Observable indicators: (a) Are orange notes factually accurate — do they correctly describe what factorisation reveals? Flag any orange notes with misconceptions (e.g., 'factorisation tells us the highest point') for correction at the start of Lesson 6. (b) Are green notes generative — do they point toward solving equations, completing the square, or the quadratic formula? These indicate readiness for Lessons 6–8. Teacher photographs the DQB after class for planning purposes.
Model Building Phase 📹 **VIDEO:** [Subtracting by Grouping (Numbers up to 10)] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12	Learners retrieve their cumulative parabola model — a hand-drawn or printed diagram of the shot put path that they have been annotating across Lessons 1–5. Today they add two new annotations: (1) They write the factorised form of the height model, −x(x − 6), alongside the expanded form $-x^2 + 6x$ already on the model. (2) Using a ruler, they mark and label the two x-intercepts on their parabola diagram: x = 0 (LAUNCH POINT) and x = 6 m (LANDING POINT), drawing dashed vertical lines from the x-	1. Say: 'Take out your parabola model. Since Lesson 1 you have been building this — today we add the most important labels yet.' 2. Direct learners to write the factorised form on their model. Circulate — check that learners write −x(x − 6), not x(x − 6) (sign matters). Correct sign errors immediately, saying: 'The negative sign is the physics — it is why the ball comes back down.' 3. Direct learners to mark x = 0 and x = 6 with dashed vertical lines. Say: 'These are called the zeros or roots of the parabola. The	Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models): The annotated parabola model is a physical artefact of growing understanding. Adding the factorised form and the zeros makes the model more precise and predictive. The 'Before/After' revision note is a metacognitive writing strategy that consolidates learning and makes conceptual change explicit — both for the learner and the teacher reviewing the models.	Observable indicators: (a) Does the learner correctly place x = 0 and x = 6 on the x-axis (not at the peak or at arbitrary points)? (b) Is the factorised form written correctly with the negative sign? (c) Does the 'Before/After' revision note show genuine conceptual change, not just a restatement of the lesson title? Teacher collects all models and uses a 3-point quick-score (0 = missing/incorrect placement; 1 = correct placement but no factorised form; 2 = correct placement, correct factorised form, meaningful revision note) to plan

🔍 Search ARES for similar readings	axis up to the curve at these points. In pairs, learners compare models and discuss: 'Does your curve cross the x-axis exactly at x = 0 and x = 6? If not, why might your hand-drawn curve be slightly off?' Pairs then write a two-sentence model revision note at the bottom of their diagram: 'Before today I thought… After today I now know…' Three volunteers share their revision notes with the class.	factorised form gave them to us instantly.' 4. Allow 5 minutes for paired comparison and the revision note. Circulate and probe: 'Does your curve cross x = 6 accurately? How could you improve your sketch?' 5. Cold-call 3 learners to read their revision notes. After each, affirm the learning and connect forward: 'You now know where the ball lands. Next lesson, we will use this to write and solve a full equation — and find the maximum height as well.' 6. Collect all models for teacher review before Lesson 6.		differentiated support for Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your planning journal:

1. FACTORISATION FLUENCY: During Phase 3 independent practice, which learners struggled with expressions where $a > 1$ (e.g., $3x^2 + 10x + 3$)? Did you have time to address sign errors in expressions with negative b terms (e.g., $x^2 - 5x + 6$)? Plan a 5-minute warm-up for Lesson 6 targeting these specific error types.
2. PHENOMENON CONNECTION: When you connected $-x(x - 6)$ to the landing point $x = 6$, did learners make the connection spontaneously, or did you need to scaffold it heavily? If heavily scaffolded, consider revisiting the Lesson 1 phenomenon video at the start of Lesson 6 and asking learners to physically point to the landing point before opening their notebooks.
3. DQB QUALITY: Review the green wonder-notes from the DQB. How many learners asked questions about finding the maximum height, the vertex, or what happens when the expression cannot be factorised easily? These are diagnostic signals for differentiation in Lessons 6–8.
4. PACING: The 80-minute lesson is dense. If Phase 2 (Observe) ran over time, consider splitting Section B ($ax^2 + bx + c$ where $a > 1$) into a separate mini-lesson or flipped pre-reading task before Lesson 6.
5. EQUITY CHECK: Were both male and female learners equally cold-called? Did learners from different counties (using the name-strip tin) get equitable airtime? Note any patterns and adjust Lesson 6 accordingly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners completed a Factorisation Investigation Worksheet in groups of 4, discovering the product-sum rule by filling a structured table and applying it to factor six quadratic expressions — including a Kericho tea-shamba area context. They then verified each factorisation by re-expanding using the FOIL method from Lesson 4. Groups recorded findings on manila sheets for gallery sharing.
What did I learn?	Learners learned that every quadratic expression of the form $x^2 + bx + c$ can be written as a product of two brackets $(x + p)(x + q)$, where $p \times q = c$ and $p + q = b$. They extended this to $ax^2 + bx + c$ ($a > 1$) using the split-middle-term and grouping method. Crucially, they discovered that factorising the shot put height model $-x^2 + 6x$ into $-x(x - 6)$ immediately reveals the two horizontal distances where height = 0 — $x = 0$ (launch) and $x = 6$ m (landing) — giving algebraic predictive power over the physical event.

How does this explain the phenomenon?	Learners explained that factorisation is the reverse of expansion: it 'un-wraps' a quadratic expression to reveal its hidden bracket structure. They connected this to the anchoring phenomenon by annotating their cumulative parabola models with the factorised form and the two x-intercepts, and wrote 'Before/After' revision notes articulating how factorisation answers the question of where the shot put lands. The Driving Question Board was updated with the insight: 'The factorised form tells us where the ball hits the ground (the zeros/roots of the expression).'

LESSON 6 (80 minutes): Expanding Products of Two Linear Brackets: Grid/Box Method and FOIL in Kenyan Contexts

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To develop learners' ability to correctly expand the product of two linear brackets into a quadratic expression using the grid/box method and FOIL, drawing on Kenyan real-life contexts, and to identify and correct common expansion errors collaboratively.
Knowledge	Learners will know that: (1) multiplying two linear expressions $(ax + b)(cx + d)$ produces a quadratic expression of the form $ax^2 + bx + c$; (2) the grid/box method organises each term multiplication systematically; (3) FOIL (First, Outer, Inner, Last) is a structured mnemonic for the same process; (4) like terms in the expansion must be collected to give the standard quadratic form.
Skills	Learners will be able to: (1) expand products of two linear brackets using the grid/box method; (2) expand products of two linear brackets using FOIL; (3) collect like terms to express the result in standard quadratic form $ax^2 + bx + c$; (4) identify and correct common errors such as sign mistakes and omission of the middle term.
Attitudes	Learners will appreciate that: (1) algebraic expansion is a precise and systematic process that rewards careful attention to signs; (2) collaborative error-correction builds mathematical confidence; (3) the same algebraic structure appears in diverse real-life situations — from garden dimensions in Nakuru to price-demand models in Mombasa — reflecting the power and universality of mathematics.
Key Inquiry Question	How do we form quadratic expressions?
Purpose in Storyline	In Lesson 5 learners identified quadratic expressions through area models and began recognising the structure $ax^2 + bx + c$. Lesson 6 now equips learners with two reliable, systematic methods — the grid/box method and FOIL — to generate (expand) quadratic expressions from the product of two linear brackets. This is a critical evidence-gathering step: every quadratic expression that will later be factorised (Lessons 7–8) or used to model the shot-put parabola's height-distance rule was first formed by expanding brackets. Mastering expansion with accuracy — including correct sign handling — is the algebraic engine that drives the rest of the unit.
Safety Notes	No physical safety hazards. Remind learners to work neatly and clearly in their exercise books so errors can be spotted and discussed. No calculator use during the expansion practice phase — the goal is procedural fluency and pattern recognition.

B. LESSON OVERVIEW

Lesson 6 is the sixth lesson in an eight-lesson evidence-gathering sequence on Quadratic Expressions and Equations. Learners engage with the expansion of products of two linear brackets using two complementary methods: the grid/box method and FOIL (First, Outer, Inner, Last). All worked examples and practice problems are drawn from Kenyan contexts — the dimensions of a school garden in Nakuru, and price-demand relationships for a fish seller at Mombasa's Kongowea market. The lesson opens with a Predict phase where learners estimate what the product of two linear expressions could look like, connecting to the curved shot-put path from the anchoring phenomenon. Learners then Observe systematic worked examples, Explain by practising and error-correcting in pairs, update the Driving Question Board, and close by revising their cumulative class model of how a quadratic expression is built. Common errors (sign errors, missing middle term, incorrect collection of like terms) are surfaced and corrected as a whole class. The lesson advances the driving question by showing learners that the rule describing the shot-put's parabolic arc — once expressed in bracket form — can be systematically expanded into the standard quadratic form they will later solve.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Graphing systems of inequalities] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two things side by side on the board: (A) the still-frame image of the shot-put parabola from Lesson 1, annotated with the placeholder rule h = (–x + 6)(x – 1) where h is height and x is horizontal distance (in metres), and (B) a rectangle drawn on the board labelled with side lengths (x + 3) cm and (x + 2) cm, described as a school garden plot in Nakuru. Learners individually write in their exercise books their predictions: 'What will the area expression of this garden look like when fully written out — and do you think it will be quadratic? Why or why not?' Learners also predict whether the shot-put rule h = (–x + 6)(x – 1) will expand into a straight-line or curved equation. After 3 minutes of silent individual prediction, learners share with a shoulder partner for 2 minutes. The teacher then takes 4–5 cold calls, recording predicted expressions on the board without affirming or correcting yet.	1. Display the two side-by-side scenarios. Say: 'Before I show you anything, I want your brain to do the work first. In Lesson 5 we saw how areas can produce quadratic shapes. Now look at this garden in Nakuru — its length is (x + 3) cm and its width is (x + 2) cm. What is the area? Write your prediction.' 2. Allow 3 minutes of silent individual think time. Do NOT circulate — let learners commit independently. 3. After 3 minutes, say: 'Now turn to your shoulder partner. Compare your predictions. Do you agree? Where do you differ?' Allow 2 minutes. 4. After partner talk, say: 'Let me hear some brave predictions.' Cold-call 5 learners by name (not volunteers). Record every prediction verbatim on the board — including incorrect ones such as 'x^2 + 5' or 'x + 5' — without comment. 5. Point to the shot-put expression h = (–x + 6)(x – 1) and ask: 'Does this look like it will give us a straight line or a curve when expanded? Think silently for 10 seconds.' [WAIT 10 SECONDS]. Cold-call 2 learners. 6. Say: 'Hold your predictions — we will return to them.' Do not erase the board predictions.	Strategy: Predict-then-compare. Learners activate prior knowledge of area and multiplication before instruction, exposing naive conceptions (e.g., 'just add the terms'). Recording all predictions publicly creates cognitive dissonance when the correct expansion is later revealed, deepening the learning moment.	Observable indicator: Do learners write an expression involving x^2 (quadratic) or only linear/constant terms? Teacher scans written predictions during cold-call phase. If fewer than half the class predicts a quadratic result, the teacher notes this and plans to make the 'two terms each multiplied by two terms' structure extra explicit during Phase 2.
Observe Phase 📹 **VIDEO:** [Graphing systems of inequalities] Source: Khan Academy (English - US curriculum)	Learners observe the teacher work through three fully structured examples on the board, each using both the grid/box method AND FOIL in parallel columns, so learners see the two methods produce identical results.	1. Before starting, say: 'I am going to show you two methods for the same process — the grid/box method and FOIL. Both give the same answer. You will choose which you prefer, but you must understand both.' 2. Work Example 1 slowly,	Strategy: Parallel method comparison with annotated worked examples. Showing both grid/box and FOIL simultaneously on the board allows learners to see structural equivalence. Colour-coded arcs for FOIL and shaded cells in the grid make each	Observable indicator: During cold-call on 8·(–3), does the learner correctly say –24? Teacher also scans copied notes — are both methods present and are signs correctly written? If learners are omitting the negative in the grid cells, the teacher pauses and re-

🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Example 1 — Garden in Nakuru (concrete, positive terms): The school garden has length (x + 3) cm and width (x + 2) cm. Find its area. Grid/box: Draw a 2×2 table. Top headers: x and +3. Side headers: x and +2. Fill in: $x \cdot x = x^2$, $x \cdot 3 = 3x$, $2 \cdot x = 2x$, $2 \cdot 3 = 6$. Collect: $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$. FOIL: First: $x \cdot x = x^2$. Outer: $x \cdot 3 = 3x$. Inner: $2 \cdot x = 2x$. Last: $2 \cdot 3 = 6$. Sum: $x^2 + 5x + 6$. Example 2 — Fish seller at Kongowea Market, Mombasa (negative constant): A fish seller's daily revenue model shows that the number of customers is (x + 8) and the price per fish in hundreds of shillings is (x – 3). Expand (x + 8)(x – 3). Grid/box: $x \cdot x = x^2$, $x \cdot (-3) = -3x$, $8 \cdot x = 8x$, $8 \cdot (-3) = -24$. Collect: $x^2 - 3x + 8x - 24 = x^2 + 5x - 24$. FOIL: First = x^2, Outer = $-3x$, Inner = $8x$, Last = -24. Sum = $x^2 + 5x - 24$. Example 3 — Shot-put parabola (negative leading coefficient, connects to phenomenon): The height of the shot-put is modelled by $h = (-x + 6)(x - 1)$. Expand. Grid/box: $-x \cdot x = -x^2$, $-x \cdot (-1) = x$, $6 \cdot x = 6x$, $6 \cdot (-1) = -6$. Collect: $-x^2 + x + 6x - 6 = -x^2 + 7x - 6$. FOIL: First = $-x^2$, Outer = x, Inner = $6x$, Last = -6. Sum = $-x^2 + 7x - 6$. Teacher explicitly says: 'This is the algebraic rule for the curve we saw in the slow-motion video.'	narrating aloud: 'I draw my box. My first expression (x + 3) goes across the top — one term per column. My second expression (x + 2) goes down the side — one term per row. Now I fill every cell.' 3. After completing the grid, say: 'Now let me show you FOIL for the same problem.' Draw an arc diagram above the expression linking F, O, I, L terms with colour-coded arcs. 4. After Example 1, ask: 'Raise your hand if your prediction on the board matches this answer.' [Acknowledge]. 'Raise your hand if your prediction was different.' [Acknowledge]. 'No shame — this is exactly why we predict first.' 5. Before Example 2, say: 'This one has a negative sign. I want everyone to watch the sign on –3 every single step.' Work through slowly. When you get to $8 \cdot (-3) = -24$, pause and say: 'Cold-call — Amina, what is 8 times negative 3?' [WAIT 10 SECONDS]. 6. After Example 2, say: 'The most common error at this step is writing +24 instead of –24. We will catch that error in our practice. 7. For Example 3, say: 'This is our shot-put. The expression came from our video. Let us decode it together.' Work through, then step back and say: '$h = -x^2 + 7x - 6$. This quadratic is the mathematical fingerprint of that parabolic arc. Every lesson from here brings us closer to using this to predict exactly where the ball lands.' 8. After all three examples, ask: 'Using only your notes, can you describe the grid/box method in one sentence to your partner?' [WAIT 15 SECONDS]. Cold-call 2	multiplication step visible. Connecting Example 3 directly to the anchoring phenomenon gives the abstract algebra immediate physical meaning.	demonstrates before moving to Phase 3.

	After each example, learners copy both methods into their exercise books. The teacher pauses after Example 2 to highlight the sign trap: 'Watch what happens with +8 and –3. This is where most errors happen.'	pairs to share.		
Explain Phase 📹 **VIDEO:** Graphing systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs on a structured 'Practise and Correct' worksheet with six problems, all set in Kenyan contexts. After every two problems, a 'Common Error Trap' is inserted — a worked example with a deliberate mistake for learners to identify and correct. Problems: (1) A rectangular maize shamba in Kericho has length (x + 7) m and width (x + 4) m. Find the area by expanding the product. (2) A mandazi seller in Kisumu charges (y + 5) shillings per mandazi and sells (y + 9) mandazi per hour. Write an expression for hourly revenue. (3) Expand (x – 4)(x + 6). (4) Expand (2x + 3)(x – 5). (5) Expand (x + 10)(x – 10). [Teacher notes this is a difference of two squares preview — do not name it yet, but let learners notice the middle term vanishes.] (6) Shot-put connection: A different student athlete throws a ball whose height is modelled by h = (x + 2)(–x + 5). Expand and write in standard form. Error Trap after Q2: 'Wanjiku expanded (x + 5)(x + 9) and got x^2 + 45. Where did she go wrong? Write the correct expansion.'	1. Distribute or display the worksheet. Say: 'Work with your partner — one person sets up the grid, the other writes the FOIL arcs. Then compare your answers. If you disagree, the grid is your referee.' 2. Circulate actively. For the first 4 minutes, focus on pairs showing sign errors on Q3 (–4)(+6). Crouch to their level and ask: 'What is negative 4 times positive 6?' [WAIT 10 SECONDS]. Do not give the answer — prompt: 'Check your sign rule.' 3. At the 10-minute mark, call attention briefly: 'Pause. Before you do Q5, predict: will the answer have a middle x-term or not? Write your prediction on the corner of your page. Then expand and check.' [WAIT 10 SECONDS for prediction]. 4. After Q5, cold-call 3 learners: 'Ochieng, what did you predict about the middle term? What did you get?' Record responses. If learners notice the middle term is zero, say: 'Interesting. File that observation — we will return to it in a future lesson.' [Do not explain difference of squares yet.] 5. For Error Trap 1 (Wanjiku): cold-call — 'Fatuma, where exactly did Wanjiku make her error?' [WAIT 15 SECONDS]. Expected answer: she multiplied only the last terms and ignored the middle cross-	Strategy: Collaborative error-correction with deliberate mistake analysis (Error Traps). By analysing Wanjiku's and Kamau's errors, learners engage in metacognitive reasoning — they must reconstruct the incorrect thinking, not just produce the right answer. This deepens procedural understanding and builds the habit of self-checking.	Observable indicators: (1) On Q5, do learners correctly predict or notice the vanishing middle term? (2) On Error Trap 1, can learners articulate the structural error (missing cross-multiplication)? (3) On Error Trap 2, can learners identify the specific step where Kamau's error occurs? Teacher uses clipboard checklist with learner names — tick if pair completes Q1–Q4 correctly, circle if sign errors persist after peer correction.

	Error Trap after Q4: 'Kamau expanded $(2x + 3)(x – 5)$ and got $2x^2 – 7x – 15$. Is he correct? Check each step using the grid method.' After all six problems, the teacher leads a whole-class error-correction discussion, projecting or writing the most common errors seen during circulation.	terms. 6. For Error Trap 2 (Kamau): cold-call — 'Mwangi, check Kamau's Outer term. What should it be?' [WAIT 10 SECONDS]. Expected: Outer = $2x\cdot(–5) = –10x$, not $–7x$. Kamau added the coefficients instead of multiplying. 7. Close Phase 3 by writing the three most common errors on the board as a class 'Watch List': (a) Sign errors with negatives, (b) Missing the middle term (adding x terms instead of producing two separate products), (c) Incorrect collection of like terms. Say: 'These three errors will cost marks. Knowing them by name means you can check for them on purpose.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Graphing systems of inequalities] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The class DQB is displayed (physical board or digital board on screen). Each pair receives two sticky notes (or writes in the DQB section of their workbook if physical resources are limited). On the first note they write one thing they can NOW explain about the shot-put parabola that they could not explain before this lesson. On the second note they write one new question this lesson has raised. Pairs post their notes to the DQB under the columns 'Now I Can Explain…' and 'New Questions We Have.' The teacher reads selected notes aloud and explicitly maps them to the driving question, then selects one 'New Question' to highlight as a bridge to Lesson 7.	1. Direct learners to the DQB. Say: 'In Lesson 1 we asked: what kind of mathematical rule could describe the curved path of the shot-put? Today we made progress. On your first sticky note, complete this sentence: Because of today's lesson, I can now explain that the expression for the shot-put's height…' [WAIT 15 SECONDS for pairs to begin writing]. 2. After 3 minutes, say: 'Now your second note — what new question has this lesson opened up? Maybe something like: if we can expand brackets, can we go in reverse? Can we un-expand? Write your question.' 3. Collect and post notes (or have pairs post themselves). Read 3–4 'Now I Can Explain' notes aloud, affirming the link to the phenomenon. 4. Select one 'New Question' — ideally one asking about reversing	Strategy: Public knowledge-building through the Driving Question Board. Making individual and pair thinking visible on a shared board creates a class knowledge artefact. Explicitly linking DQB entries to the driving question reinforces that each lesson is not isolated — it is one piece of evidence in a larger explanatory argument about the parabolic phenomenon.	Observable indicator: Are the 'Now I Can Explain' notes mathematically accurate? Specifically, do learners mention expanding, quadratic form, or the connection to the curve? Notes that are vague ('I understand more') signal that the learner has not yet crystallised the algebraic insight — teacher flags these pairs for follow-up in Lesson 7.

		the process (factorising) — and say: 'This question — [read it] — is exactly where Lesson 7 takes us. Hold that thought.' 5. Update the class 'Evidence We Have Gathered' column on the DQB: write 'Lesson 6: Expanding brackets → quadratic expressions. The shot-put height h = (–x + 6)(x – 1) expands to h = –x² + 7x – 6.'		
Model Building Phase 📹 **VIDEO:** Graphing systems of inequalities Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative 'Class Model' — a large annotated diagram that has been built across Lessons 1–5. The model currently shows: the parabolic arc with labelled key points; a table of (distance, height) values; the observation that the relationship is not linear; and the identification of the expression structure ax² + bx + c from Lesson 5. In Lesson 6, learners now add a new layer: 'HOW a quadratic expression is formed — the expansion process.' Each learner sketches the grid/box for (–x + 6)(x – 1) in their workbook model section and writes the full expansion showing each step. They annotate with arrows: 'Two linear factors → expansion → standard quadratic form.' The teacher then poses a final whole-class model-building challenge: 'If we know the expanded form of the shot-put expression is h = –x² + 7x – 6, and we want to find where the ball hits the ground, what must we do next?' Learners write their ideas in their model — this seeds the concept of solving a quadratic equation (Lesson 8).	1. Display the cumulative class model. Say: 'Our model has been growing since Lesson 1. Let us add today's layer.' 2. Instruct: 'In your model section, draw the grid/box for (–x + 6)(x – 1). Show every step. Label it: Expansion Process — Grid/Box Method.' Allow 5 minutes. 3. Circulate and look for learners who are setting up the grid incorrectly (wrong headers). Quietly correct by pointing to the Observe phase notes. 4. After 5 minutes, ask: 'Who can tell me — in the model, what does each cell of the grid represent?' Cold-call 2 learners. [WAIT 10 SECONDS each]. Expected: each cell is the product of one term from each bracket. 5. Now ask the forward-looking question: 'The ball lands when h = 0. So we need –x² + 7x – 6 = 0. We cannot solve this the way we solved linear equations. What might we need to do?' [WAIT 15 SECONDS]. Accept all ideas without evaluating. Record them on the model under 'Open Questions.' 6. Close by saying: 'In Lesson 7 we learn to reverse the expansion — to factorise. And in Lesson 8 we use that to solve equations like this one and predict exactly where the	Strategy: Cumulative model revision with forward-bridging. By physically adding the expansion process to the growing class model, learners see their algebraic knowledge as a progressive, connected structure rather than isolated procedures. The forward-bridging question ('how do we find where h = 0?') creates productive intellectual tension that motivates the next two lessons.	Observable indicator: In the model sketch, do learners correctly show all four products in the grid and correctly collect –x² + 7x – 6? Do the annotations connect expansion back to the parabolic path? Exit check: teacher asks 3 learners to read their 'Open Questions' entry aloud — are they framing a question about finding where h = 0, or about reversing expansion? These responses guide the framing of Lesson 7's opening.

		ball lands. You are building toward a complete answer to our driving question.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) How many pairs made persistent sign errors even after the Error Trap discussion? If more than one-third of the class, plan a 5-minute warm-up at the start of Lesson 7 using a 'sign error detective' routine before introducing factorisation. (2) Did learners notice the vanishing middle term in Q5 (the difference of two squares preview)? If so, note which learners articulated it clearly — they may serve as peer explainers when this pattern is formalised later in Grade 10. (3) Were the DQB 'New Questions' focused on reversing expansion (factorising)? If learners instead asked about the graph of the quadratic, consider briefly acknowledging the graph connection and linking it to the phenomenon before Lesson 7. (4) How was the pacing across the two methods (grid and FOIL)? If FOIL caused more confusion than grid, emphasise grid as the primary method in Lesson 7 practice, with FOIL as a checking tool. (5) Were the Kenyan contexts (Nakuru garden, Kongowea fish seller, Kericho shamba) meaningful to learners, or did they feel forced? Adjust future contexts based on learner engagement signals observed during Phase 3 circulation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed the teacher work through three fully annotated examples of bracket expansion using both the grid/box method and FOIL in parallel, with Kenyan contexts (Nakuru school garden, Kongowea fish seller, shot-put parabola). They observed that $(-x + 6)(x - 1)$ expands to $-x^2 + 7x - 6$ — the algebraic rule for the shot-put's curved path.
What did I learn?	Learners learned that the product of two linear expressions $(ax + b)(cx + d)$ always produces a quadratic expression; that the grid/box method organises all four term-products systematically; that FOIL (First, Outer, Inner, Last) provides an equivalent structured procedure; and that like terms must be collected carefully — especially when negative signs are involved — to arrive at the standard form $ax^2 + bx + c$.
How does this explain the phenomenon?	Learners explained and corrected common expansion errors (missing middle term, sign errors with negatives, incorrect collection of like terms) through structured pair practice and Error Trap analysis. They updated the class Driving Question Board and cumulative model to show that the shot-put's parabolic height rule $h = (-x + 6)(x - 1)$ is mathematically equivalent to $h = -x^2 + 7x - 6$, bringing the class closer to being able to predict where the ball lands.

LESSON 7 (80 minutes): Forming and Solving Quadratic Equations in Real-Life Situations	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To form and solve quadratic equations arising from real-life situations — including the parabolic path of a thrown object and the dimensions of a farmer's shamba — using factorisation and the quadratic formula.
Knowledge	Learners know that a quadratic equation is of the form $ax^2 + bx + c = 0$ and can be solved by factorisation or by applying the quadratic formula $x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$. They understand that the solutions (roots) represent meaningful real-world values such as landing distance or shamba dimensions.
Skills	Learners can: (i) translate a real-life word problem into a quadratic equation in standard form; (ii) solve quadratic equations by factorisation; (iii) solve quadratic equations using the quadratic formula; (iv) interpret both roots in context and reject non-physical solutions.
Attitudes	Learners appreciate that algebra is a powerful tool for predicting real outcomes — from where a shot put will land on the sports field to how a Kericho tea farmer can maximise the area of a shamba — and develop confidence in tackling multi-step problems.
Key Inquiry Question	How do we form and solve quadratic equations in real-life situations?
Purpose in Storyline	In earlier lessons learners formed quadratic expressions (Lessons 1–3), factorised them using common factor, grouping, difference of two squares, and the trinomial method (Lessons 4–6). Now, in Lesson 7, learners take the decisive step: they set a quadratic expression equal to a value, rearrange into standard form, and solve — directly answering the driving question 'exactly where will the ball land?' and 'what dimensions give the farmer the required area?' This is the algebraic climax of the evidence-gathering sequence before the final Model-Building synthesis lesson.
Safety Notes	No physical hazards. Ensure learners do not use mobile phones for social media during the lesson; devices (if available) are for GeoGebra simulation or calculator use only. Learners working in pairs should manage shared graph paper carefully to avoid disputes.

B. LESSON OVERVIEW
Learners are presented with two real-life problems anchored to the unit phenomenon: (1) the height equation of a shot put thrown on the school sports field, $h = -x^2 + 6x$ (where h is height in metres and x is horizontal distance in metres), and (2) a Kericho tea farmer who wants to fence a rectangular shamba of area 40 m² with one side 3 m longer than the other. Through a structured Predict–Observe–Explain–DQB–Model sequence, learners form quadratic equations in standard form, solve them by factorisation and by the quadratic formula, interpret both roots, and reject non-physical solutions. They update the class Driving Question Board and revise their cumulative parabolic-path model to include the landing point of the shot put.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Multiplication as	Learners are shown a projected still-frame of the shot put parabola from the anchoring phenomenon	Display the parabola still-frame and equation on the board. Say: 'Look at the curve we have been	Anchored Prediction — by committing to a numerical estimate before instruction, learners	Observable indicator: learner has written a specific numerical prediction AND a one-sentence

[scaling with fractions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	with the height equation $h = -x^2 + 6x$ written beneath it. Working individually for 3 minutes, each learner writes a prediction in their exercise book: 'At what horizontal distance x does the ball hit the ground (h = 0)? Give a numerical estimate and explain your reasoning in one sentence.' They then share predictions with a shoulder partner for 1 minute before three cold-call responses are taken.	studying since Sports Day. The equation that describes the height of the shot put is $h = -x^2 + 6x$. The ball starts at ground level, rises, then comes back down. Your job right now — before we do any algebra — is to predict: at what distance x does the ball hit the ground again?' Allow 3 minutes individual silent thinking (enforce WAIT TIME). Circulate and scan books for common misconceptions — watch for learners who say x = 6 by inspection versus those who say x = 3 (confusing vertex with landing). After pair-share, cold-call exactly 3 learners: one who estimated a small value, one a larger value, one who attempted algebra. For each, ask: 'What reasoning led you there?' Record all three predictions on the board under the heading 'OUR PREDICTIONS — we will test these.'	activate prior knowledge of what h = 0 means physically (the ball is on the ground) and create cognitive dissonance that motivates the algebraic method to follow. Predictions are displayed publicly to be revisited, building metacognitive awareness.	reason in their book. Teacher scans for: (a) learners who correctly identify that h = 0 is the condition (strong prior knowledge); (b) learners who guess randomly (need scaffolding in Phase 3). Note names of both groups for targeted questioning.
Observe Phase 📹 **VIDEO:** [Multiplication as scaling with fractions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs on a structured evidence-gathering worksheet containing two problems presented side by side. Problem A (Shot Put — Sports Field, Nairobi): The height h (metres) of a shot put above the ground at horizontal distance x (metres) is given by $h = -x^2 + 6x$. Set h = 0 and record the equation. Problem B (Shamba — Kericho Tea Farm): A farmer in Kericho wants to fence a rectangular plot. One side is 3 m longer than the other. The area must be 40 m². Let the shorter side = x metres. Write an expression for the area and set it equal to 40. Pairs spend 8 minutes forming both equations in standard form ($ax^2 + bx + c = 0$), recording every algebraic step. After 8 minutes, two pairs join to	Distribute the worksheet. Say: 'Your task in Phase 2 is purely to set up the equations — do not solve yet. I want to see every step written down. For Problem A, ask yourself: what does h = 0 mean in the real world? For Problem B, draw a rectangle in your book, label the sides, and write the area formula first.' Circulate for 8 minutes. At the 4-minute mark, pause the class and ask one cold-call: 'Fatuma, read me your equation for Problem A — just the equation, not the solution.' Correct any errors in forming standard form. At the 8-minute mark, say: 'Now compare with your group of four — do all four equations look the same? If not, find the disagreement before I come to your table.' After 3 minutes group	Concrete-to-Abstract Bridging — learners first draw a physical representation (rectangle for the shamba, parabola arc for the shot put) before writing algebra, ensuring the equation is grounded in a geometric or physical meaning. Side-by-side problems reveal that different real-life contexts produce the same algebraic structure (quadratic in standard form), building transferable pattern recognition.	Observable indicator: both equations correctly written in standard form ($ax^2 + bx + c = 0$) with all steps shown. Teacher specifically checks: (a) sign handling when rearranging $h = -x^2 + 6x$ to $x^2 - 6x = 0$ (multiplication by −1); (b) expansion of $x(x + 3) = 40$ to $x^2 + 3x - 40 = 0$. Learners who cannot form standard form are flagged for targeted support in Phase 3.

	form a group of four to compare their standard-form equations before whole-class check.	comparison, conduct whole-class check: write correct standard forms on the board: Problem A: $x^2 - 6x = 0$, Problem B: $x^2 + 3x - 40 = 0$. Ask: 'Raise your hand if your equation for Problem B matches what is on the board.' Address discrepancies publicly.		
Explain Phase 📹 **VIDEO:** Multiplication as scaling with fractions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners solve both equations using two methods, guided by teacher-led worked examples followed by independent practice. Method 1 — Factorisation: Learners solve $x^2 - 6x = 0$ by factorising as $x(x - 6) = 0$, giving $x = 0$ or $x = 6$. They interpret both roots: $x = 0$ is where the ball was thrown (start), $x = 6$ m is where it lands. They solve $x^2 + 3x - 40 = 0$ by finding two numbers that multiply to −40 and add to 3 (−5 and 8), factorising as $(x - 5)(x + 8) = 0$, giving $x = 5$ or $x = -8$; they reject $x = -8$ because a length cannot be negative, so the shorter side is 5 m and the longer side is 8 m. Method 2 — Quadratic Formula: Learners re-solve $x^2 + 3x - 40 = 0$ using $x = (-b \pm \sqrt{b^2-4ac}) / 2a$ with a=1, b=3, c=−40, confirming $x = 5$ and $x = -8$. They record a summary rule: 'Always interpret BOTH roots in context and reject any root that is not physically possible.' Pairs then tackle two extension problems: (i) a shot put where $h = -2x^2 + 8x$ — find where it lands; (ii) a rectangular maize farm in Rift Valley where one side is 2 m longer than double the other side and area = 60 m^2 — find dimensions.	Begin with a 10-minute teacher-led demonstration at the board, narrating every step aloud. For $x^2 - 6x = 0$ say: 'I notice there is a common factor of x — this is exactly what we practised in Lesson 4. I factorise to get $x(x - 6) = 0$. Now the Zero Product Property tells me: if two things multiply to zero, at least one of them must be zero. So either $x = 0$ or $x - 6 = 0$. Go back to the phenomenon — what does $x = 0$ mean on the sports field? And what does $x = 6$ mean?' Allow 10 seconds WAIT TIME, then cold-call: 'Kipchoge, what does $x = 6$ tell us about the shot put?' For $x^2 + 3x - 40 = 0$, model the factor-search aloud: 'I need two integers that multiply to −40 and add to +3. I will list factor pairs of 40: 1×40, 2×20, 4×10, 5×8. Which pair has a difference of 3? Five and eight. Since the product is negative, one must be negative. Does −5 + 8 = 3? Yes. So $(x - 5)(x + 8) = 0$.' After demonstrating the quadratic formula re-solve, say: 'Both methods give $x = 5$ and $x = -8$. Now I must ask — can a side of a shamba be −8 metres? Discuss with your partner for 30 seconds.' Cold-call: 'Amina, what do we do with $x = -8$ and why?' During paired extension practice, circulate and prompt struggling learners with: 'What is your value of a, b, and c? Write them down before	Worked Example Analysis followed by Faded Scaffolding — the teacher models full solutions with explicit narration of reasoning (not just procedure), then learners tackle near-transfer problems (same structure, different numbers/context) with decreasing teacher support. The 'interpret both roots' rule is explicitly named and posted on the board as a sensemaking anchor, preventing purely procedural application. Connecting $x = 0$ and $x = 6$ back to the shot put parabola on the board links the algebra to the visible phenomenon throughout.	Observable indicators: (a) learner correctly identifies and rejects non-physical root with a written reason ('a length cannot be negative'); (b) learner correctly substitutes into the quadratic formula with correct sign for b; (c) learner's extension problem shows dimensions that multiply to the stated area (verification step). Teacher uses a quick show-of-whiteboards (or open books) after the extension problems: 'Hold up your answer for the dimensions of the maize farm.' Correct answer: sides are 5 m and 12 m for problem (ii). Learners who show incorrect answers are identified for the DQB phase questioning.

		substituting.' After 15 minutes, select one pair to present extension problem (ii) on the board while the class checks their own work.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Multiplication as scaling with fractions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or draws two boxes in their book if notes are unavailable). On the first note they write one thing the lesson has helped them understand about the driving question — specifically connecting today's algebra to the shot put or the shamba. On the second note they write one question that still puzzles them or a new question the lesson has raised (e.g., 'What if the equation has no real roots — does that mean the ball never lands?'; 'How do we find the highest point of the parabola, not just where it lands?'). Notes are placed on the class DQB under two columns: 'NOW WE CAN EXPLAIN…' and 'WE STILL WONDER…'. The teacher reads three 'wonder' notes aloud and the class votes on which question is most important to investigate in Lesson 8.	Say: 'We have been building our Driving Question Board since the first lesson. Today we answered a big part of the driving question — we can now predict exactly where the shot put lands using algebra. Take two minutes to write your sticky notes.' Enforce silence for 2 minutes. As learners post notes, quickly group similar 'wonder' questions on the board. Read three aloud: choose one that anticipates the vertex/maximum point (connects to Lesson 8 model building), one that questions the discriminant (b^2-4ac), and one that connects to the shamba context. Say: 'These three questions are going to guide our final lesson. Which one do you think we absolutely must answer to complete our model of the shot put path?' Hands vote. Record the winning question as the Lesson 8 focus on the board.	Public Knowledge Building (DQB Protocol) — making individual understanding and curiosity visible to the whole class creates a shared intellectual record of the unit's progress. The 'NOW WE CAN / WE STILL WONDER' structure reinforces that science and mathematics are iterative, and that solved questions generate new questions. Voting on the Lesson 8 focus gives learners agency over the inquiry direction.	Observable indicator: learner's 'NOW WE CAN EXPLAIN' note makes a specific, accurate claim connecting today's algebra to the phenomenon (not a vague statement). Teacher reads every note quickly during posting and flags any that reveal persistent misconceptions (e.g., a learner who writes '$x = 6$ is the highest point' — this confuses landing distance with the vertex and must be corrected before Lesson 8).
Model Building Phase 📹 **VIDEO:** [Multiplication as scaling with fractions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Finding a Linear Product Using a Quadratic (PLIX)]	Learners retrieve their cumulative parabolic-path model diagram — a hand-drawn or printed arc of the shot put that they have annotated across Lessons 1–6 with labels for the expression, the factored form, and key features. Today they add two new annotations: (1) they mark and label the landing point at $x = 6$ m on the horizontal axis with the equation $x(x - 6) = 0 \rightarrow x = 0$ or $x = 6$; (2) they draw and label the shamba rectangle inset in the corner of their model page showing sides 5 m and 8 m with the equation $(x - 5)(x + 8) = 0$.	Distribute (or ask learners to open) their cumulative model pages. Say: 'Your model has been growing every lesson. Today you are adding the most important piece yet — the actual landing point of the shot put. Mark $x = 6$ on your diagram right now.' Circulate for 2 minutes ensuring all learners correctly place the landing point on the horizontal axis (not the highest point). Say: 'Now add the shamba rectangle as an inset — this shows that the same algebra works for completely different real-life problems.' After 5 minutes	Cumulative Model Revision — by physically annotating the same diagram across all eight lessons, learners construct a tangible record of conceptual growth. Each addition is not a new page but a deepening of the same representation, making the unit's coherence visible. The closing return to Phase 1 predictions is a metacognitive bookend: learners explicitly compare their pre-instruction intuition with their post-instruction understanding, consolidating learning.	Observable indicators: (a) landing point marked at $x = 6$ on horizontal axis (not at $x = 3$, the vertex); (b) shamba rectangle labelled with correct dimensions 5 m × 8 m; (c) Model Statement explicitly names both solution methods and the rejection criterion for non-physical roots. Teacher collects three model pages at random at the end of the lesson as an exit artefact for written feedback before Lesson 8.

Source: CK-12 🔍 Search ARES for similar readings	They write a 'Model Statement' beneath the diagram: 'A quadratic equation h = 0 or Area = k can be solved by factorisation or the quadratic formula to find real-world values like landing distance or shamba dimensions. We reject solutions that are not physically meaningful.' Pairs compare model statements and agree on wording before writing the final version.	annotation, say: 'Write your Model Statement. It should answer, in two sentences, how quadratic equations connect to real-life prediction. Compare with your partner — do your statements agree on what we reject non-physical roots?' Cold-call one pair: 'Wanjiku and Otieno, read your model statement to the class.' Class gives a thumbs-up/thumbs-down on accuracy. Correct any errors. Close the lesson by returning to the original prediction board from Phase 1: 'Look at our predictions from the start of the lesson. Who predicted x = 6? What helped you — or what do you now understand that you didn't then?'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Could learners independently set up the quadratic equation in standard form from a word problem, or did they need prompting — if many struggled with Problem B (the shamba), consider adding a 'draw and label first' scaffold as a permanent routine in Lesson 8. (2) Did learners consistently interpret both roots and reject non-physical solutions, or did they stop at the first positive root — if the latter, the Zero Product Property and its contextual meaning need reinforcement. (3) Were the extension problems (Rift Valley maize farm, $h = -2x^2 + 8x$) appropriately challenging for faster learners, or was more differentiation needed? (4) Review the three model pages collected: do the Model Statements demonstrate genuine conceptual understanding (connecting algebra to prediction) or procedural description only? Use findings to adjust the Lesson 8 model-building synthesis, particularly the focus on the vertex/maximum point that learners voted onto the DQB as their priority wonder question.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that setting h = 0 in the shot put equation $h = -x^2 + 6x$ produces a quadratic equation $x^2 - 6x = 0$, and that a real-world area constraint on a Kericho shamba similarly produces $x^2 + 3x - 40 = 0$ — demonstrating that the same algebraic structure emerges from very different physical situations.
What did I learn?	Learners learned to form quadratic equations in standard form from real-life word problems, to solve them by factorisation (common factor and trinomial) and by the quadratic formula, and to interpret both roots in context — rejecting non-physical solutions such as negative lengths. They established that x = 6 m is the landing distance of the shot put and that the shamba dimensions are 5 m × 8 m.
How does this explain the phenomenon?	Learners explained that the parabolic path of the shot put can be decoded algebraically: by setting the height equation equal to zero and solving, we predict exactly where the ball lands before it lands. The quadratic formula provides a universal method when factorisation is not obvious. A solution that is mathematically valid but physically impossible (negative dimension) must be rejected

	by interpreting the root in real-world context.

LESSON 8 (80 minutes): Final Explanation: Reversing the Expansion — Factorisation, Difference of Two Squares, and the Full Picture of the Parabolic Path

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate the entire unit by applying factorisation — including the difference of two squares identity — to expressions that arose from the tiled courtyard and Kericho shamba contexts, and to use these tools to give a complete, evidence-based explanation of the parabolic path of the shot put thrown on sports day.
Knowledge	Learners correctly state and apply the difference of two squares identity $(a+b)(a-b) = a^2-b^2$; reverse the process of expansion to factorise quadratic expressions of the forms x^2+bx+c and ax^2+bx+c; recognise difference of two squares as a special factorisation case; and connect factorisation to solving quadratic equations that model real contexts.
Skills	Learners systematically factorise quadratic expressions using appropriate methods; apply the difference of two squares identity to simplify and solve; form and solve quadratic equations from the shamba and shot-put contexts; and revise their cumulative model to produce a complete, annotated explanation of the parabolic phenomenon.
Attitudes	Learners appreciate that algebra is a reversible, flexible tool — expansion and factorisation are two directions of the same mathematical road — and demonstrate confidence in selecting the most efficient method for a given expression.
Key Inquiry Question	How do we form and solve quadratic equations in real-life situations?
Purpose in Storyline	This is the final lesson of the unit. Learners have expanded expressions, derived identities from area, and solved equations. Today they reverse the expansion process — factorising expressions that grew out of the courtyard tiles and the Kericho shamba — and use factorisation as the final key to fully answering the driving question: predicting exactly where the shot put lands. The DQB is completed, and the cumulative model reaches its final form.
Safety Notes	No physical safety concerns. Remind learners to show all working steps clearly to avoid algebraic sign errors, especially when applying the difference of two squares with negative terms.

B. LESSON OVERVIEW

Lesson 8 is the capstone lesson for Sub-Strand 1.3. The lesson opens by returning to the anchoring phenomenon — the shot put parabola from Nairobi sports day — and asking learners to predict, before instruction, whether factorisation can help find where the ball lands. In the Observe phase, learners work with two revisited supporting phenomena: (1) the tiled courtyard, whose area expression $x^2+7x+10$ is now reversed via factorisation, and (2) the Kericho shamba, whose boundary conditions generate a difference-of-two-squares expression. In the Explain phase, learners consolidate all three factorisation methods (common factor, trinomial, difference of two squares) and apply them to solve quadratic equations. The Driving Question Board is formally completed as a class, with learners writing final answers to questions they posted in earlier lessons. The Model Building phase culminates in learners annotating their full parabola model — now labelled with roots, vertex region, and the algebraic rule — and comparing it to their Lesson 1 initial sketch. The lesson closes with a structured Final Explanation protocol in which each learner writes a three-part evidence-based answer to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a still image projected on the board: the parabolic arc of the shot put from sports day, with two points marked on the ground labelled A (launch point) and B (landing point) and the equation $h = -x^2 + 6x$ printed beneath it, where h is height in metres and x is horizontal distance in metres. Working individually for 3 minutes, each learner writes answers to two questions in their exercise books: (1) 'Without solving, predict: at what horizontal distances is the height of the ball equal to zero — i.e., where does the ball touch the ground?' and (2) 'What mathematical operation might help you find those distances from the equation $h = -x^2 + 6x$?' Learners then share predictions in pairs (1 minute) before the teacher takes 4 cold-call responses from different corners of the room.	Display the shot-put arc image and equation prominently. Say: 'We have been studying this parabola since Lesson 1. Today is the day we fully decode it. Before I show you anything new, I want your prediction.' Allow 3 minutes of individual silent writing — enforce quiet. After pair-share, cold-call exactly 4 learners: one from the front-left, one from the back-right, one from the middle, one who has been quieter this unit. For each response, wait 12 seconds after posing the question before accepting an answer. Record all four predictions verbatim on the board under the heading 'Our Predictions — we will check these by the end of class.' Do NOT confirm or correct yet. Say: 'Every one of these ideas is on trial today. We will use evidence to decide which prediction is correct.'	Predict-before-instruction anchoring: By committing predictions to paper before instruction begins, learners activate prior knowledge from Lessons 1–7 (expansion, identities, solving by inspection) and create a personal stake in resolving the cognitive dissonance. This sets factorisation up as a necessary tool — not an arbitrary exercise — because learners genuinely want to know who predicted correctly.	Teacher circulates during individual writing and notes: (a) Do learners attempt to substitute h=0? (b) Do any learners write $x(-x+6)=0$ or attempt factorisation already? (c) Are predictions numerical (e.g., 'x=0 and x=6') or only verbal? These observations inform the pace of Phase 3. If fewer than 3 learners attempt any algebraic manipulation, slow down the Explain phase and add a bridging step.
Observe Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners rotate through two structured evidence stations (15 minutes each, groups of 4–5). STATION A — THE TILED COURTYARD REVISITED: Groups receive a printed card showing a rectangular courtyard whose total tiled area is described as $x^2+7x+10$ square metres. The card states: 'In Lesson 2 you found this area by multiplying two side lengths. Today, work backwards: if the area is $x^2+7x+10$, what must the two side lengths be?' Groups write all factor-pair combinations of 10 (i.e., 1×10, 2×5) and check which pair sums to 7. They then write the factorised form and sketch the rectangle with labelled sides (x+2) and (x+5), confirming	Before stations begin, say: 'You are not getting new information today — you are reversing a process you already know. Think of it like ugali: you already know how to mix the flour and water to get ugali. Today you are figuring out what flour and water were used from the ugali that is already cooked.' Distribute station cards. Circulate every 3–4 minutes, spending equal time at both stations. At Station A, prompt stuck groups: 'What two numbers multiply to give 10? List every pair.' Wait 10 seconds. Then: 'Now which of those pairs adds to 7?' At Station B, if groups do not recognise difference of two squares, ask: 'What is special	Structured evidence stations with a context-reversal design: each station presents a context learners have already encountered (courtyard from Lesson 2, shamba from Lesson 4) but inverts the cognitive direction — from product back to factors. This 'cognitive reversal' strategy makes factorisation feel like detective work rather than a new algorithm, grounding it in familiar phenomena and strengthening the connection between area models and algebraic structure.	Collect one written record per group from each station at the end of the rotation. Check: (a) At Station A, does the factorised form $(x+2)(x+5)$ appear with a verification step (re-expansion)? (b) At Station B, does the group correctly identify the expression as a difference of two squares and write $(x+9)(x-9) = 40$ leading to $x^2-81=40$, then $x^2=121$, $x=\pm11$? Groups who reach only part (a) or (b) of Station B are prioritised for teacher support in Phase 3.

	area by re-expanding. STATION B — THE KERICHO SHAMBA: Groups receive a card: 'A farmer in Kericho fences a rectangular shamba. The length is $(x+9)$ metres and the width is $(x-9)$ metres. (a) Expand to find the area expression. (b) Identify what type of expression this is. (c) Factorise it back using the identity you recognise. (d) If the area of the shamba is 40 m², form and solve the equation to find x.' Groups record observations on the card and then in their exercise books. After both stations, a 5-minute gallery share: one reporter per group stands by their station work and explains to a passing group.	about the two brackets — look at the numbers inside them very carefully.' Wait 12 seconds. Do not give the answer; redirect to their Lesson 5 notes on the $(a+b)(a-b)$ identity. During the gallery share, listen for the key phrase 'difference of two squares' — note which groups use it and which do not; this informs cold-calling in Phase 3.		
Explain Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher leads a 25-minute structured whole-class explanation organised in three tiers. TIER 1 — CONSOLIDATING FACTORISATION METHODS (8 min): The teacher writes three expressions on the board drawn from the unit's contexts: (i) $3x^2+9x$ (from the courtyard border — common factor), (ii) $x^2+7x+10$ (courtyard area — trinomial factorisation), (iii) x^2-81 (shamba with 9m offset — difference of two squares). Learners work each one in their books for 90 seconds, then the teacher cold-calls one learner per expression to write their working on the board. Class checks each step. TIER 2 — DIFFERENCE OF TWO SQUARES AS A SPECIAL IDENTITY (7 min): Teacher formally states: 'The expression a^2-b^2 is special. It factors as $(a+b)(a-b)$. This is not a coincidence — you proved it in Lesson 5 using area rectangles.' Teacher writes	For Tier 1 cold-calls, select learners strategically: choose the learner who struggled most visibly at Station A for expression (ii) — public success builds confidence. For Tier 2, after writing each example, say: 'You have 45 seconds — go' and time it visibly. After the neighbour-check say: 'Put your hand up if your answer matches your neighbour's.' Scan the room; note mismatches and address the most common error on the board. For Tier 3, before factorising the shot-put equation, pause and say: 'I want you to try this yourself before I write anything. Set h=0. You have 90 seconds.' Walk the room during this time. Wait the full 90 seconds before asking for a volunteer — do not shorten this wait. When revealing x=0 and x=6, draw a quick arc on the board with the two roots labelled and say: 'Does this shape remind you of anything we have been looking at since Lesson 1?' Pause 10 seconds for the	Tiered consolidation with a phenomenon payoff: The three-tier structure moves from procedural fluency (Tier 1) through conceptual depth (Tier 2) to phenomenon resolution (Tier 3). By reserving the shot-put solution for Tier 3, the explain phase has a built-in narrative arc — learners feel the algebraic tools they have been practising 'clicking into place' at the moment of maximum interest. This is the 'tool earns its purpose' sensemaking strategy.	During Tier 1 board work, observe whether the selected learners correctly handle sign changes (especially −3 as a common factor). In Tier 2, the neighbour-check provides a real-time scan — if more than 30% of pairs are mismatched on any example, pause and reteach that case before moving on. In Tier 3, circulate during the 90-second independent attempt: if fewer than half the class writes $-x(x-6)=0$ correctly, work through the common-factor step explicitly before proceeding to the solution.

	three rapid-fire examples with Kenyan contexts: (a) 'A maize plot of area x^2-25: what are the dimensions?' (b) 'A term in the shot-put height formula: $9-x^2$ — factorise it.' (c) 'Simplify $(x+7)(x-7)$ without expanding fully.' For each, learners write answers individually (45 seconds) before a neighbour-check. TIER 3 — RETURNING TO THE SHOT PUT: FULL SOLUTION (10 min): The teacher writes $h = -x^2+6x$ on the board. 'This is the equation from our prediction phase. Set $h=0$ because we want to know when the ball is at ground level.' Learners factorise $-x^2+6x=0$ step by step: take out common factor $-x$ to get $-x(x-6)=0$, so $x=0$ or $x=6$. Teacher connects: '$x=0$ is where the athlete stands. $x=6$ metres is where the ball lands. Our factorisation just told us something the athlete could not calculate in their head — exactly where the ball lands.' Teacher returns to the four predictions on the board and checks them against $x=0$ and $x=6$ with a show-of-hands celebration for correct predictors.	visual connection to the parabola to land.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12	The full DQB (displayed since Lesson 1) is now brought to the front of the room or projected. Each group receives a set of green sticky notes (representing 'answered') and orange sticky notes (representing 'partially answered — needs more study'). Groups have 8 minutes to review every question card on the DQB that belongs to their group and decide: Is this question now answerable using what we have learned? For each 'yes', they write a concise evidence-based answer on a green note and attach it over	Before distributing sticky notes say: 'In Lesson 1 you posted questions because something puzzled you. Today you have the tools to answer those questions. A good scientist or mathematician does not just solve problems — they close the loop and record what they now understand.' Give groups exactly 8 minutes — use a visible countdown timer. During group work, circulate and prompt: 'Can you answer this question using factorisation? Can you answer it using the shot-put equation?' For the whole-class	DQB closure protocol: The act of physically converting question cards to answer cards provides visible, tangible evidence of knowledge growth. The contrast between the blank/question-laden board from Lesson 1 and the green-covered board in Lesson 8 is a powerful metacognitive tool — learners literally see their own learning. The distinction between green and orange notes also models scientific honesty: not everything is fully resolved, and that is acceptable.	Count the ratio of green to orange notes per group. Any group with more than 2 orange notes on foundational questions (forming expressions, factorising trinomials, solving equations) is flagged for follow-up practice in the next class session or homework. Collect one green note per group as an exit artefact — check whether answers reference evidence from the unit (area models, shot-put equation, shamba dimensions) rather than stating only procedures.

🔍 Search ARES for similar readings	the original question. For each 'partially answered', they write what they now know and what still remains on an orange note. The teacher then conducts a 5-minute whole-class DQB review: walking along the board, reading one green note per group aloud, and inviting the class to agree or refine the answer. Special attention is given to the unit's three KICD Key Inquiry Questions — the teacher ensures these are explicitly answered on the board before closing.	review, cold-call a different learner from each group to read their green note — aim for 5–6 green notes across the class. For the three KICD Key Inquiry Questions, ask the whole class: 'Who can state the answer to this question in one sentence?' Wait 12 seconds before accepting. Write the agreed sentence on the board in a different colour as the 'class consensus answer.' Close by saying: 'Every question you posted has been met with evidence. That is what mathematics looks like when it is working.'		
Model Building Phase 📹 **VIDEO:** Polynomials intro Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens to their cumulative model page (started in Lesson 1 and revised in Lessons 3, 5, and 7). They have 10 minutes to complete the Final Model, which must include: (1) a labelled parabolic arc showing the shot-put path with $x=0$ (launch) and $x=6$ m (landing) marked as roots on the horizontal axis; (2) the equation $h=-x^2+6x$ written on the arc; (3) the factorised form $-x(x-6)=0$ shown beside the roots; (4) an annotation connecting the shamba and courtyard contexts: 'The same factorisation tools we used for the Kericho shamba and the tiled courtyard also decoded the shot-put path.' After completing the model, each learner writes a Final Explanation in their exercise books responding to the driving question: 'How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems — from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho?' The Final Explanation must have exactly three parts labelled: (A)	Distribute the Final Model checklist (four required elements listed above) as a printed slip or written on the board. Say: 'Your model today is not a new drawing — it is the completed version of everything you have been building. Every lesson added one more piece. Today you add the last piece and step back to see the whole picture.' Circulate during model completion; prompt learners who leave the equation off their arc: 'What rule governs this curve? Write it on the arc.' For the Final Explanation writing, say: 'Write in full sentences. Pretend you are explaining this to a Form 1 student who has never seen a quadratic expression. If they can understand your explanation, you have truly understood it yourself.' Allow the full 10 minutes without interruption. In the last 2 minutes, cold-call 3 learners to read their 'biggest change' sentence aloud. Use these to close the lesson: 'From a puzzling curved path on sports day to a complete algebraic explanation — that journey is what this unit has been about. Well	Final model comparison and three-part written explanation: The side-by-side comparison of Lesson 1 and Lesson 8 models activates metacognitive reflection — learners see their own conceptual growth concretely. The three-part written explanation (Form, Factorise, Solve) mirrors the three KICD Key Inquiry Questions and forces learners to integrate procedural and conceptual knowledge into a coherent narrative. Writing to a hypothetical Form 1 audience (the 'Explain It to a Younger Learner' strategy) requires genuine comprehension rather than rote reproduction of steps.	Collect all Final Explanation books at the end of class. Assess each against three criteria: (1) Does Part A correctly describe forming a quadratic expression with a real-life example? (2) Does Part B name at least two factorisation methods and apply one correctly with working shown? (3) Does Part C connect factorisation to a real numerical answer (e.g., $x=6$ metres for the shot put, or specific shamba dimensions)? Learners who meet all three criteria have demonstrated mastery of the sub-strand SLOs. Learners meeting only one or two criteria are identified for targeted consolidation in the next lesson or through peer tutoring pairs.

	What I can now FORM — how to write a quadratic expression or equation from a real situation; (B) What I can now FACTORISE — how to reverse expansion to find structure; (C) What I can now SOLVE — how factorisation gives real answers (dimensions, distances, landing points). Learners compare their Lesson 1 initial sketch to their final model and write one sentence: 'The biggest change in my model from Lesson 1 to Lesson 8 is…'	done.'		

D. TEACHER REFLECTION

After this lesson, reflect on the following questions: (1) When learners factorised the shot-put equation $-x^2+6x=0$ independently in Tier 3, what percentage of the class correctly extracted the common factor without prompting? If fewer than 60% succeeded independently, the unit needs an additional practice session on common-factor extraction before this final lesson is taught again. (2) Did the DQB closure feel genuine or procedural? Were learners engaged in deciding green vs. orange, or were they simply filling in notes? If the latter, consider introducing a 'challenge round' where one group questions another group's green-note answer. (3) Compare the Lesson 1 initial sketches with the Lesson 8 final models: do learners' models show qualitative improvement in mathematical labelling (roots, equation, factorised form), or only in artistic detail? Mathematical labelling is the indicator that matters. (4) Did the Kenyan contexts (shamba, courtyard, shot put on sports day) feel connected throughout the unit, or did they feel like separate examples? If learners' Final Explanations treat each context as isolated, future planning should build more explicit cross-context references in earlier lessons. (5) Which of the three KICD Key Inquiry Questions generated the most confident whole-class responses on the DQB? Which generated the most orange notes? Use this to identify which aspect of the sub-strand (forming, factorising, or solving) needs the most reinforcement before summative assessment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that the area expression $x^2+7x+10$ from the tiled courtyard — which they had previously generated by expanding $(x+2)(x+5)$ — could be reversed: starting from the expanded form and recovering the two side-length factors $(x+2)$ and $(x+5)$ by finding a pair of numbers that multiply to 10 and add to 7. Learners also observed that the Kericho shamba's area expression $(x+9)(x-9)$ produced x^2-81, and that this 'difference of two squares' structure allowed instant factorisation using the identity $(a+b)(a-b) = a^2-b^2$. Finally, learners observed that the shot-put height equation $h=-x^2+6x$, when set equal to zero, could be factorised as $-x(x-6)=0$, yielding $x=0$ and $x=6$ — the exact horizontal distances at which the ball is at ground level.
What did I learn?	Learners learned that factorisation is the reverse of expansion: every quadratic expression that arose from an area context was originally a product of two factors, and algebraic reasoning can recover those factors. They learned three factorisation methods: (1) extracting a common factor, (2) factorising a trinomial x^2+bx+c by finding two numbers that multiply to c and sum to b, and (3) applying the difference of two squares identity $a^2-b^2=(a+b)(a-b)$ as a special, efficient case. They learned that setting a factorised quadratic equation equal to zero and applying the zero-product property yields the roots — the values of the unknown that solve the equation — and that these roots carry real-world meaning: landing distances, shamba dimensions, courtyard side lengths.
How does this explain the phenomenon?	The parabolic path of the shot put thrown on sports day in Nairobi is described by a quadratic equation. Quadratic expressions arise naturally from area — whenever two variable lengths are multiplied, the result is a quadratic. Factorisation reverses this

	process, breaking a quadratic expression back into the two lengths that produced it. When the height equation of the shot put is set to zero and factorised, the roots reveal exactly where the ball meets the ground. The difference of two squares identity is a powerful special case that arises when two lengths differ only in sign — as in the Kericho shamba with sides $(x+9)$ and $(x-9)$. Across all these contexts — a tiled courtyard, a Rift Valley shamba, and an athlete's throw — the same algebraic structure (the quadratic) and the same algebraic tool (factorisation) explain and predict real outcomes. The driving question is fully answered: quadratic expressions and equations are the mathematical language of curved, symmetric, real-world phenomena.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.